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A Tale of Two Anomalies: The Implications of Investor Attention for Price and Earnings Momentum*

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Abstract

We study how investor attention affects two well-known stock market anomalies: price momentum and earnings momentum. Using a comprehensive set of investor attention measures, we find a striking contrast. Stocks that attract more attention show stronger price momentum but weaker earnings momentum. This suggests attention plays two opposing roles: when investors pay less attention and ignore earnings news, stock prices underreact, leading to stronger earnings momentum; but when attention is high, behavioral biases intensify and fuel overreaction-driven price momentum. We also find that different types of attention—from institutions versus retail investors, and active versus passive attention—have distinct effects.

Keywords: Attention, Price Momentum, Earnings Momentum, Anomalies, Overreaction, Underreaction
JEL Classification Codes: G12, G14, G40

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1. Introduction

Human attention is limited, and decades of research show we can only process so much information at once (Kahneman, 1973).[1] This limitation fundamentally shapes how investors learn from and respond to information in financial markets. Traditional theories assume markets quickly incorporate new information, but growing evidence suggests investor attention systematically affects behavior and prices.[2] This paper shows that investor attention plays opposing roles in two major market anomalies: price momentum (when past winners keep winning and losers keep losing) and earnings momentum (when stocks with positive earnings surprises continue to outperform and those with negative earnings surprises continue to underperform). We find that attention weakens earnings momentum but amplifies price momentum.

This hypothesis stems from theories linking earnings momentum to underreaction and price momentum to overreaction. When investors pay insufficient attention to earnings news, they do not fully process its implications, causing initial underreaction (Ball and Brown 1968; Bernard and Thomas 1989; Hirshleifer, Lim, and Teoh 2011). Prices gradually adjust as information spreads, creating earnings momentum. Following rational inattention models (Sims 2003; Peng 2005; Peng and Xiong 2006), we expect higher attention enables more complete information processing, reducing earnings momentum.

Conversely, behavioral biases like overconfidence and excessive extrapolation generate the overreaction underlying price momentum (e.g., De Long, Shleifer, Summers, and Waldmann, 1990; Daniel, Hirshleifer, and Subrahmanyam, 1998; Barberis, Greenwood, Jin, and Shleifer, 2015; Da, Huang, and Jin, 2021; Bose, Cordes, Nolte, Schneider, and Camerer, 2022; Afrouzi, Kwon, Landier, Ma, and Thesmar, 2023). We argue attention is necessary for these biases

[1] See Pashler and Johnston (1998) and Koch, Poljac, Müller, and Kiesel (2018) for a review of these studies.

[2] See, e.g., Merton (1987), Gervais, Kaniel, and Mingelgrin (2001), Huberman and Regev (2001), Hirshleifer and Teoh (2003), Grullon, Kanatas, and Weston (2004), Hou and Moskowitz (2005), Peng (2005), Peng and Xiong (2006), Hong, Torous, and Valkanov (2007), Seasholes and Wu (2007), Cohen and Frazzini (2008), Corwin and Coughenour (2008), Barber and Odean (2008), DellaVigna and Pollet (2009), Hirshleifer, Lim, and Teoh (2009), Da, Engelberg, and Gao (2011), Fang, Peress, and Zheng (2014), Lou (2014), Kacperczyk, Van Nieuwerburgh, and Veldkamp (2016), Ben-Rephael, Da, and Israelsen (2017), Kaniel and Parham (2017), Gargano and Rossi (2018), Ben-Rephael, Carlin, Da, and Israelsen (2021), Chen, Tang, Yao, and Zhou (2022), and Chen, An, Wang, and Yu (2023).

to affect behavior and trading, so higher attention strengthens price momentum. Intuitively, behavioral biases can influence prices only when investors notice, process, and act on relevant signals, making attention a natural conditioning variable for their market impact.

While price and earnings momentum overlap—price movements reflect both fundamentals and behavioral forces—they remain empirically distinct, with correlations between measures ranging from 0.172 to 0.349. Our goal is to examine how attention interacts differently with underreaction and overreaction mechanisms. We also analyze orthogonalized versions of these momentum measures to isolate their distinct components. This unified framework reveals that attention's impact depends critically on the source of return predictability: it dampens anomalies driven by inattention but amplifies those driven by behavioral biases.

To test this, we examine widely used attention proxies plus recent measures, organized along two dimensions: active versus passive attention, and investor sophistication.

Active (top-down) attention reflects deliberate information-seeking—through searches, social media activities, or trading. Passive (bottom-up) attention reflects information supply or exogenous factors not requiring deliberate action. Sophisticated investors likely process information analytically, while unsophisticated investors may exhibit more behavioral biases.

Our proxies include: *Turnover* (average daily trading volume), capturing active attention from all investor types (e.g., Barber and Odean, 2008; Gervais, Kaniel, and Mingelgrin, 2001; Cosemans and Frehen, 2021; Barber, Huang, Odean, and Schwarz, 2022); *Bloomberg AIA* (Ben-Rephael et al., 2017) and *Edgar Downloads* (Lee, Ma, and Wang, 2015), measuring active information acquisition by primarily institutional and sophisticated retail investors; social media measures including *Social Attention* (Cookson et al., 2024a), *StockTwits* (Cookson and Niessner, 2020), and *Tweets* (Chen and Hwang, 2022), plus *Google ASV* (Da et al., 2011), capturing active retail attention; and passive measures including *News Articles* and *–Distraction* (Kempf, Manconi, and Spalt, 2017), capturing reduced attention when a firm's large institutional shareholders experience shocks in their unrelated stock holdings. We also construct a *Composite Attention* measure averaging quintile ranks across all proxies.

For momentum, we use past 6- or 11-month returns for price momentum (Jegadeesh and Titman, 1993), and three earnings measures: standardized unexpected earnings (*SUE*) (e.g., Foster, Olsen, and Shevlin, 1984; Chordia and Shivakumar, 2006), announcement abnormal

returns (*ABR*), and analyst forecast revisions (*FREV*) (Chan, Jegadeesh, and Lakonishok, 1996). We aggregate these into composite measures and analyze both raw and orthogonalized versions.

Using portfolio sorts, we find two key results with *Composite Attention*. First, price momentum profits are substantially higher for high-attention stocks—exceeding low-attention stocks by 47–74 basis points (bps) monthly across different factor models. Second, earnings momentum profits are consistently stronger for low-attention stocks by 99-119 bps monthly. These patterns hold across individual momentum measures.[3]

With orthogonalized measures isolating distinct components, the contrast sharpens: attention increases price momentum alphas by 64–91 bps (up from 47–74 unorthogonalized) while earnings momentum effects remain stable. This strongly supports our hypothesis.

Individual attention proxies reveal heterogeneous patterns. Active attention from all investors (*Turnover*) strengthens price momentum and weakens earnings momentum. Institutional attention (*Bloomberg AIA*) reduces earnings momentum without amplifying price momentum, consistent with efficient fundamental processing. *Edgar Downloads* shows similar patterns but also positively relates to price momentum, suggesting sophisticated investors may exhibit behavioral biases or correlate with unsophisticated investor attention.

Social media attention strongly amplifies price momentum and reduces earnings momentum. The price momentum effect aligns with our hypothesis that retail attention amplifies behavioral biases and is consistent with Cookson et al. (2024a) finding that social media attention predicts negative returns. However, the negative earnings momentum relationship suggests that social media participants are not all "dumb money"—some contribute to price discovery around earnings (Chen et al., 2014; Cookson and Niessner, 2020).

Results based on *Google ASV* show earnings momentum is significant for low-search stocks but not high-search stocks, though differences are not statistically significant. We find no significant relationship with price momentum.

Among passive attention measures, news coverage (*News Articles*) attenuates earnings

[3]As one would expect, the newer factor models are better at explaining the unconditional price and earnings momentum anomalies compared to older factor models like the Fama and French (1993) three-factor model. Despite the increased explanatory power of these benchmark models, our finding that attention interacts with price and earnings momentum anomalies in opposite ways remains robust even when we compute abnormal returns using these more stringent models.

momentum without amplifying price momentum, suggesting it aids efficient incorporation of earnings information. *–Distraction* correlates negatively with other attention measures and shows sensitivity to price momentum crashes without robust relationships to either anomaly.

Additional tests using cross-sectional regressions, independent sorts, and longer holding periods confirm these results: price momentum reversals are stronger for high-attention stocks (consistent with amplified overreaction), while earnings momentum shows little reversal (consistent with underreaction).

In summary, investor attention acts as a double-edged sword: amplifying price momentum while dampening earnings momentum. This supports the view that attention alleviates underreaction to fundamentals but exacerbates behavioral biases. The effects vary by investor sophistication—sophisticated investors primarily correct earnings underreaction, improving efficiency, while less sophisticated investors amplify price overreaction. Notably, even retail attention helps alleviate earnings underreaction, suggesting that when directed toward fundamental news, it can facilitate price discovery.

Our study contributes to several strands of the literature. Prior research shows prices underreact when investors are distracted (DellaVigna and Pollet, 2007; Hirshleifer et al., 2009; Ben-Rephael et al., 2017), while increased retail attention associates with short-term overvaluation (Barber and Odean, 2008; Da et al., 2011).[4] By classifying attention measures by channel and investor type, we provide insights into their distinct roles in both underreaction- and overreaction-based anomalies. We are also the first to systematically examine how social media attention relates jointly to price and earnings momentum.

We also shed new light on the relationship between price and earnings momentum. Chordia and Shivakumar (2006) find substantial overlap between them, but we show they are not identical—they exhibit opposite patterns with attention, suggesting attention shapes their divergence.[5]

[4]Market-wide investor attention has also been linked to lower future market returns (Yuan, 2015; Chen, Tang, Yao, and Zhou, 2022; Cookson, Lu, Mullins, and Niessner, 2025), with Da, Hua, Hung, and Peng (2026) showing that this predictive effect differs between institutional and retail investors.

[5]While the literature has extensively documented that earnings momentum is primarily an underreaction phenomenon, Bathke, Mason, and Morton (2019) report evidence of reversals for a sample of firms with uncorrelated seasonal earnings changes, which they interpret as evidence of overreaction to earnings news. As we discuss in Section 6.3, this reversal pattern is likely driven by the overlap between price and earnings momentum, as there is little evidence of reversals once earnings momentum is orthogonalized with respect

Finally, we contribute to the anomaly literature. While underreaction and overreaction have been studied extensively but separately,[6] recent work explores common drivers like sentiment (Stambaugh, Yu, and Yuan, 2012) and news (Engelberg, McLean, and Pontiff, 2018). Han, Huang, Huang, and Zhou (2022) find that mispricing is concentrated in high-volume stocks, while Jiang, Liu, Peng, and Wang (2022) find that the profits of multiple anomalies in China are higher after spikes in social media activity. While these studies find anomaly profits generally rise with their proposed factors, we show attention affects overreaction- and underreaction-based anomalies in opposite directions, with effects varying by attention channel and investor sophistication. This suggests no single common factor fully explains the cross-section of anomalies.

The paper is organized as follows. Section 2 presents our attention-based hypothesis. Section 3 describes the data. Section 4 tests how composite investor attention affects price and earnings momentum. Section 5 examines individual attention proxies. Section 6 reports additional results on cross-sectional regressions, price momentum crashes, reversals, and firm size. Section 7 concludes.

2. Hypothesis development

We propose that investor attention plays two distinct—and seemingly contradictory—roles in how markets process information. First, when attention is limited, investors may overlook important information about a firm's value. Greater attention helps investors process this information more thoroughly, reducing underreaction. Second, when investors do pay attention, their behavioral biases—such as overconfidence and the tendency to extrapolate past trends—can lead them to overreact to signals like past returns or their own private information. In this case, more attention amplifies the overreaction.

to price momentum.

[6]See Hirshleifer (2001) and Barberis and Thaler (2003) for reviews of related behavioral theories. Empirically, De Bondt and Thaler (1985) examine stock market overreaction by testing whether extreme movements reverse, while Zarowin (1989) investigates overreaction to extreme earnings news. Hong, Lim, and Stein (2000) study how the slow diffusion of firm-specific news (i.e., underreaction) affects price momentum (see also Hong and Stein, 1999). Jackson and Johnson (2006) explore whether underreaction serves as a unifying explanation for stock market anomalies, and Stambaugh, Yu, and Yuan (2012) investigate whether overvaluation is more pronounced during high sentiment periods when optimistic investors dominate. For a review on the anomaly literature, see Hou, Xue, and Zhang (2020).

Theory and evidence suggest that price momentum and earnings momentum correspond to these two distinct types of mispricing. Limited attention has long been identified as a key reason markets underreact to earnings announcements, which represent fundamental news about firm value (Ball and Brown 1968; Bernard and Thomas 1989; Hirshleifer et al. 2011). Rational inattention models formalize this idea by treating limited attention as a constraint on investors' ability to process value-relevant information (Peng 2005, Peng and Xiong 2006, Hirshleifer et al. 2011). When investors pay little attention to a firm, they fail to fully absorb the implications of its earnings news, causing prices to initially underreact. Over time, as this information gradually gets incorporated, prices drift in the direction of the earnings surprise—creating earnings momentum. This framework predicts that greater attention should reduce this underreaction, weakening earnings momentum for stocks that attract more investor interest.[7]

Price momentum, by contrast, is commonly attributed to overreaction driven by behavioral biases. When investors extrapolate past returns too far into the future (De Long et al. 1990), or when overconfidence in their private signals gets reinforced by recent price movements through self-attribution (Daniel et al. 1998), they overreact to past returns or private information, generating price momentum. Since attention is necessary for investors to form and act on these biased beliefs, we expect that higher attention intensifies these biases. Thus, unlike earnings momentum, increased attention should strengthen price momentum.

Our framework suggests that attention's effect on market efficiency depends on what drives investor reactions to a given signal. For anomalies rooted in underreaction to fundamental information—like earnings announcements—attention helps overcome processing constraints and reduces mispricing. For anomalies triggered by overreaction to past price trends or related private signals, attention can amplify the overreaction. This dual role leads to our central hypothesis:

Hypothesis: *Stocks with higher investor attention should exhibit stronger price momentum, while stocks with lower investor attention should exhibit stronger earnings momentum.*

In formulating this hypothesis, we remain agnostic about the specific information in-

[7]This form of underreaction, driven by information processing constraints, differs from the bias-based underreaction described by Barberis et al. (1998), which they attribute to investor conservatism.

vestors focus on. When attention is low, investors likely overlook all relevant information—including earnings news—leading to stronger earnings momentum. When attention is high, investors may overreact to whatever information they do attend to, including the stock's recent price history and their own private signals, thereby strengthening price momentum. Note that price momentum need not arise solely from overreaction to past price movements; it can also stem from overreaction to investors' private information, as Daniel, Hirshleifer, and Subrahmanyam (1998) describe.

3. Data and variable description

Our sample consists of CRSP common stocks with share codes of 10 or 11 (ordinary shares) from January 1967 to December 2022. Monthly stock returns are adjusted for delisting following Shumway (1997).

3.1. Attention proxies

Since investor attention cannot be directly observed, we examine nine proxies proposed in the literature. For each proxy, we use average values over the previous 12 months to reduce transitory noise, though our results are similar using shorter windows (e.g., three months; see Table IA.1).

3.1.1. Active attention measures

We first describe the active attention measures. All measures are monthly averages computed over the past 12 months. For measures available at the daily level, we first compute monthly aggregates or averages, then take the average of these monthly values over the past 12 months.

The first proxy is *Turnover*, defined as the average of monthly daily share turnover over the past 12 months. Daily turnover equals the number of shares traded divided by the number of shares outstanding at the end of the day. We follow Gao and Ritter (2010) to adjust Nasdaq trading volume for comparability with NYSE/AMEX volume. Trading volume captures active investor attention because investors cannot trade a stock without

paying attention to it, and when they do attend, heterogeneous opinions generated by biased information processing can trigger more trading (Odean, 1998; Scheinkman and Xiong, 2003). The link between trading volume and investor attention is well-supported empirically.[8] One advantage of trading volume is its availability across a large cross-section of firms over an extended period, making it appealing for large-scale asset pricing studies.

The next two measures are *Bloomberg AIA* (abnormal institutional attention), based on Bloomberg terminal readership activities, and *Edgar Downloads*, based on downloads from the SEC's EDGAR system. *Bloomberg AIA* is the average daily maximum readership (DMR) for a firm in a given month, capturing institutional investor attention (Ben-Rephael et al. 2017).[9] The measure covers 2010–2021 with an average of 2,276 firms per month.

Edgar Downloads counts monthly downloads of a company's SEC filings.[10] Previous studies suggest these downloads reflect information acquisition by institutional and sophisticated retail investors (see, e.g., Lee, Ma, and Wang, 2015; Chen, Cohen, Gurun, Lou, and Malloy, 2020; Li and Sun, 2022). This measure covers 2003–2017 with an average of 2,987 firms per month.

The next three measures are social media-based. *Social Attention* is the first principal component of attention from StockTwits, X, and Seeking Alpha (Cookson et al. 2024a). We take daily averages for each month, covering an average of 700 firms from 2012–2022. We also examine two individual platforms. *StockTwits* is the monthly StockTwit message volume for a stock from 2009–2022 (Cookson and Niessner 2020). *Tweets* is the monthly count of tweets and retweets about a stock on X (formerly Twitter) from 2012–2019 (Chen and Hwang 2022).[11]

[8]Lo and Wang (2000) show that trading volume tends to be higher for larger firms that attract more investor attention. Chordia and Swaminathan (2000) show that even controlling for size, high-volume firms respond more quickly to market information than low-volume firms. Gervais et al. (2001) show that increases in trading volume raise a firm's visibility and attract more attention. Similarly, Barber and Odean (2008) and Cosemans and Frehen (2021) use abnormal trading volume to capture investor attention.

[9]Bloomberg records hourly user activities (search and readership) for a given firm relative to the previous 30 days. Daily (DMR) equals zero, one, two, three, or four if the maximum hourly activity falls below the 80th percentile, between the 80th and 90th, between the 90th and 94th, between the 94th and 96th, or greater than the 96th percentile of the historical distribution, respectively.

[10]We thank James Ryans for providing the data from Lawrence, Ryans, and Sun (2017), where following Loughran and McDonald (2017), IP addresses exceeding 50 daily downloads are filtered as robots. Robots are removed since they skew download counts and their automated downloads are less likely to reflect active investor analysis.

[11]We thank the authors of Cookson et al. (2024a), Cookson and Niessner (2020), and Chen and Hwang

We also construct *Google ASV* (Google abnormal search volume), defined as the percentage change between Google's daily Search Volume Index (SVI) for a stock ticker and its past six-month median, averaged monthly.[12] Da et al. (2011) propose this measure and present evidence that it likely captures attention from less sophisticated retail investors.

3.1.2. Passive attention measures

We examine two passive attention measures proposed in prior literature. The first is *News Articles*, defined as the monthly count of RavenPack news articles mentioning a firm (see, e.g., Kolasinski et al., 2013; Kelley and Tetlock, 2017), averaged over the past 12 months. This measure captures the supply of information and, since investors are more likely to attend to firms mentioned in the media (Barber and Odean, 2008), reflects supply-driven attention from both sophisticated and retail readers.

The second measure, *–Distraction*, is the negative of Kempf et al. (2017)'s distraction measure, averaged over the last four quarters. It is constructed from institutional investor portfolio holdings filed on Form 13F with the SEC. For a given stock, "distraction" periods occur when an investor's holdings in unrelated industries experience extreme (highest or lowest) returns across the 12 Fama–French industries. During these periods, attention-constrained institutional investors likely shift attention toward those industries and away from the focal firm (Corwin and Coughenour, 2008). The distraction effect's strength for a given investor depends on the portfolio weight of the affected industry.

The firm-level distraction measure aggregates across the firm's institutional shareholders, assigning greater weight to investors who own larger fractions of the firm's shares and for whom the firm represents a larger portfolio share.

3.1.3. Composite attention measures

Individual attention proxies can be noisy and have inconsistent sample coverage. To construct a more robust and comprehensive measure, we combine information across all available proxies. At the beginning of each month, we sort firms into quintiles based on each individ-

(2022) for sharing these measures.

[12] We exclude tickers with generic meanings, such as "GPS" for Gap Inc.

ual attention proxy. *Composite Attention* is then defined as the average of a firm's quintile ranks across all available proxies for that month. This approach maximizes both time-series and cross-sectional coverage.

3.1.4. Summary statistics of attention measures

Table 1 reports summary statistics and correlation coefficients for attention measures in Panels A and B, respectively. Values shown are time-series averages of statistics computed for each cross-section. Average daily *Turnover* is 0.5% with a standard deviation of 1%. Daily *Bloomberg AIA* has a mean of 0.086 and standard deviation of 0.094. Average monthly *Edgar Downloads* is 447 with a standard deviation of 891. For social media measures, average daily *Social Attention* is –0.029 with a standard deviation of 0.488, average monthly *StockTwits* is 130.2 with a standard deviation of 980.9, and average monthly *Tweets* is 309.4 with a standard deviation of 1,536.7. Average daily *Google ASV* is 0.040 with a standard deviation of 0.09. Average monthly *News Articles* is 90.2 with a standard deviation of 502.9. The mean of *–Distraction* is -0.157 with a standard deviation of 0.035.

Panel B of Table 1 presents quintile rank correlations for all attention measures. *Turnover* shows its highest correlations with social media-based proxies: *Tweets* (0.67) and *StockTwits* (0.65), followed by *Edgar Downloads* (0.58) and *News Articles* (0.56). Correlations are moderate with *Bloomberg AIA* (0.37) and lower with *Social Attention* (0.12) and *Google ASV* (0.05).

Among active attention measures, *Bloomberg AIA*, *Edgar Downloads*, and the three social media proxies exhibit strong correlations with each other (ranging from 0.43 to 0.77) and are highly correlated with *News Articles*. *Google ASV* has consistently low correlations with other measures, ranging from 0.02 to 0.13. Notably, correlations between *–Distraction* and all other proxies are generally negative, indicating it captures a different dimension of investor attention. Finally, *Composite Attention* is highly correlated with all nine attention proxies, confirming its effectiveness in aggregating information from individual measures.

3.2. Price and earnings momentum measures

Next, we detail the construction of our price and earnings momentum measures.

3.2.1. Price momentum

We use two measures to capture price momentum (Jegadeesh and Titman 1993). The first, *Past 6-mth Ret*, is the cumulative return from month $t-7$ to month $t-2$. The second measure, *Past 11-mth Ret*, is the cumulative return from month $t-12$ to month $t-2$. We skip month $t-1$ to reduce microstructure noise (see, for example, Fama and French, 1996). These measures are available from January 1967 to December 2022 for an average of 4,013 firms per month, as reported in Panel A of Table 2.

We also construct a *Composite Price Momentum* measure by averaging the prior month's quintile ranks of available individual price momentum measures. Panel A of Table 2 reports rank correlations between composite and individual measures. Correlations among the two individual price momentum measures and between individual and composite measures are high, at 0.68 and 0.92, respectively.

3.2.2. Earnings momentum

We measure earnings momentum using three approaches based on year-to-year changes in realized earnings, stock price reactions, and analysts' earnings forecasts, respectively.

First, standardized unexpected earnings (SUE) for a firm in month t equals the firm's most recently announced quarterly earnings (Compustat item IBQ) minus earnings four quarters prior, normalized by the standard deviation of earnings changes over the previous eight quarters. We require non-missing quarterly earnings announcement dates (Compustat item RDQ) between months $t-6$ and $t-1$. This measure is available from September 1971 to December 2022 for an average of 3,648 firms per month, as reported in Panel B of Table 2.

Second, ABR is the cumulative abnormal stock return from two trading days before to one trading day after the most recent earnings announcement. Abnormal returns are computed by subtracting CRSP value-weighted market index returns, and we require the earnings quarter-end month to fall within six months prior to portfolio formation (Chan

et al. 1996). This measure is available from July 1971 to December 2022 for an average of 3,702 firms per month.

Third, we use *FREV* from Chen and Zimmermann (2022), defined as the six-month moving average of past monthly price-scaled changes in analysts' FY1 consensus earnings forecasts. The consensus forecasts come from the *meanest* item in the Institutional Brokers Estimate System (I/B/E/S) Summary U.S. file. This measure is available from September 1976 to December 2022 for an average of 2,712 firms per month.

We also construct a *Composite Earnings Momentum* measure by averaging the prior month's quintile ranks of available individual earnings momentum measures. Panel B of Table 2 reports rank correlations between composite and individual earnings momentum measures. Correlations among the three earnings momentum measures range from 0.12 to 0.34, indicating they capture distinct dimensions of earnings momentum: realized fundamental changes, immediate stock market reactions, and analysts' belief updating.

3.2.3. Correlation between price and earnings momentum measures

Our empirical analysis tests whether investor attention attenuates underreaction-based earnings momentum while amplifying overreaction-related price momentum. At the same time, the two anomalies are likely positively correlated because earnings news drives stock returns and forward-looking stock returns summarize news about future fundamentals like cash flows and earnings. Empirically, Chordia and Shivakumar (2006) find that an earnings momentum factor largely explains price momentum profits.

However, Table 2 shows that correlation between composite price momentum and composite earnings momentum is only moderate, at 0.349 (Panel A). Correlations between *Composite Price Momentum* and individual earnings momentum measures are similar, ranging from 0.17 to 0.32 (Panel B). These moderate correlations suggest that despite overlap between the two measure sets, substantial independent variation remains in each.

To more cleanly isolate what drives these differences, we construct orthogonalized versions of the two composite momentum measures. At the beginning of each month, we estimate a cross-sectional regression of *Composite Price Momentum* on *Composite Earnings Momentum* and use the residual to obtain an orthogonalized price momentum measure purged of earnings

momentum's confounding effect. Similarly, we form an orthogonalized composite earnings momentum measure purged of price momentum's effect.

4. Composite attention and momentum anomalies

We begin our analysis by examining how composite investor attention moderates the profitability of price and earnings momentum portfolios. Our empirical strategy relies on two-way portfolio sorts based on attention and momentum measures. We first present results for individual momentum strategies and then turn to results based on composite momentum measures.

Each month t, using data through the end of month $t-1$, we sort all firms into quintiles based on an attention measure. Within each attention quintile, we then sort firms into quintiles based on either a price momentum or earnings momentum measure. We compute equal-weighted returns for these double-sorted portfolios in month t, holding portfolios for one month and rebalancing monthly. The return spread between the top and bottom momentum quintiles measures momentum profit within each attention group.

4.1. Composite attention and individual momentum portfolios

Table 3 presents results from the two-way portfolio sorting analysis, reporting one-month-ahead long–short portfolio returns for price momentum strategies across quintiles of *Composite Attention*. The two panels correspond to price momentum portfolios formed with *Past 6-mth Ret* and *Past 11-mth Ret*, respectively.

Panel A, based on *Past 6-mth Ret*, reports price momentum profits across the five *Composite Attention* quintiles. The reported portfolio performance metrics are excess returns, defined as portfolio returns (R_p) in excess of the risk-free rate (R_f) as proxied by the one-month Treasury Bill rate, the q-4 alpha (Hou, Xue, and Zhang, 2015), the q-5 alpha (Hou, Mo, Xue, and Zhang, 2021), the six-factor (FF6) alpha (Fama and French, 2018), and the DHS alpha (Daniel, Hirshleifer, and Sun, 2020). The two q-factor models are included because they subsume a wide range of anomalies. We include the FF6 model because it controls for price momentum through the up-minus-down (UMD) factor, and the DHS model

because it controls for earnings momentum through the post earnings-announcement drift factor (PEAD) factor. Factor returns are downloaded from the respective authors' websites and cover our full sample period, except for DHS factors, which begin in July 1972.

Across specifications, price momentum profits are higher among stocks with greater investor attention, consistent with our main hypothesis. The difference in momentum profits between the high- and low-attention quintiles is positive and statistically significant in most cases, ranging from 44 bps per month (DHS alpha) to 105 bps per month (excess return).

Panel B, based on *Past 11-mth Ret*, similarly finds that price momentum profits are larger in the high-attention quintile than in the low-attention quintile, with statistically significant differences in three out of five specifications.

We next turn to earnings momentum with Table 4 reporting the results. In Panel A, using *SUE* as the earnings momentum proxy, the differences in earnings momentum profits between the high- and low-attention quintiles are all negative and significant, ranging from –57 bps per month (excess return) to –101 bps per month (q-5 alpha). Panels B and C, which use *ABR* and *FREV*, respectively, find similar patterns. Importantly, earnings momentum profits decline with attention, in stark contrast to the price momentum results, providing support for the dual role of investor attention.

4.2. Composite attention and composite momentum portfolios

The results for individual momentum measures suggest that investor attention reduces earnings momentum profits while increasing price momentum profits. This naturally raises the question of whether attention operates on the common components of the price and earnings momentum strategies, respectively. To address this, firms are first sorted into quintiles based on *Composite Attention* and then sorted into quintiles based on either *Composite Price Momentum* or *Composite Earnings Momentum* within each attention quintile.

Table 5 reports the results. The differences in momentum profits between high- and low-attention quintiles are economically large, statistically significant, and sharply distinct across the two momentum strategies. For *Composite Price Momentum*, the High–Low attention differences are 111, 74, 51, 57, and 47 bps per month in excess returns, q-4, q-5, FF6, and DHS alphas, respectively. For *Composite Earnings Momentum* profits, the corresponding

differences are –66, –99, –111, –92, and –119 bps per month in the respective profits, all highly significant. The alpha differences are actually stronger than the excess return differences because the factor models better explain earnings momentum profits of the high-attention quintile than those of the low-attention quintile.[13]

Panel C reports the difference between Panels A and B for the Low, High, and High–Low attention quintiles. This comparison provides a joint test of the opposing effects of attention on price and earnings momentum. If attention simultaneously amplifies price momentum profits but dampens earnings momentum profits, then combining the two strategies should yield large and significant profits for the High–Low attention spread. Consistent with this prediction, the High–Low attention differential is both economically sizable and statistically significant across all specifications. Overall, these results provide support for our attention hypothesis.

As noted earlier, the composite price and earnings momentum measures are positively correlated, with a correlation of 0.349, indicating overlapping information content. This overlap may obscure the distinct roles of attention if price momentum partly reflects delayed reactions to earnings news, or if earnings momentum partly reflects overreaction to price signals correlated with earnings. To better isolate overreaction-related price momentum from underreaction-based earnings momentum, we construct orthogonalized versions of the composite momentum measures and use the orthogonalized momentum measures for all subsequent analyses unless otherwise mentioned.

Table 6 reports the results using these orthogonalized measures. The differences in orthogonalized momentum profits between high- and low-attention quintiles tend to be statistically significant and larger than those observed for the unorthogonalized measures.

Specifically, the differences in orthogonalized composite price momentum profits between the high- and low-attention quintiles are 108 bps per month in excess returns and 64–91 bps per month in factor alphas, compared with corresponding values of 111 bps and 47–74 bps in Table 5.[14]

[13]These new factor models have been shown to fully explain earnings momentum in value-weighted portfolios, e.g., the DHS model. In equal-weighted portfolios, we see that they explain earnings momentum better only in the high-attention quintile.

[14]One question regarding Panel A of Table 6 is why risk-adjusted price momentum profits are negative among low-attention stocks, and why these negative profits primarily drive the return differences between

Similarly, the differences in orthogonalized composite earnings momentum profits between the high- and low-attention quintiles are –87 bps per month in excess returns and –98 to –113 bps per month in factor alphas, compared with –66 bps and –92 to –119 bps, respectively, in Table 5. In Panel C, the High–Low attention spread continues to exhibit large and statistically significant differences between price and earnings momentum. These results further reinforce the conclusion that investor attention plays fundamentally different roles in overreaction-based and underreaction-based momentum anomalies.

5. Individual attention measures and momentum anomalies

Having presented results based on *Composite Attention*, we dive deeper into the individual attention proxies, which differ along two dimensions: whether they capture active versus passive attention, and whether they primarily reflect the attention of more versus less sophisticated investors.

We refer to active attention (top-down) measures as those that reflect conscious focus and deliberate actions toward a stock. Accordingly, we classify trading-based measures and measures that capture active information acquisition or social media activity as active attention. Conversely, passive attention (bottom-up) measures capture variation in the information environment or exogenous conditions that can attract attention without requiring deliberate action by investors.

In addition, the role of attention may depend on investor sophistication. Institutional investors tend to be more sophisticated than retail investors, and there can also be heterogeneity in sophistication within each group. Attention from sophisticated investors is more likely to improve decision making and price efficiency, whereas attention from less sophisticated investors, who may be more prone to behavioral biases, can contribute to mispricing. Examining heterogeneity across attention proxies helps identify which channels drive our

high- and low-attention stocks. We attribute this pattern to two related forces: (i) the ability of more recent factor models to explain price momentum returns, and (ii) the impact of momentum crashes. We discuss these two mechanisms in more detail in Section 6.2.

results and sheds light on how different forms of attention, generated by different investor clienteles, shape price and earnings momentum.

Below, we investigate how the different types of attention contribute to price and earnings momentum, respectively. As in earlier sections, we conduct two-way sorted portfolio analyses by sorting firms into quintiles based on an attention proxy and then, within each attention quintile, sorting firms into quintiles based on the orthogonalized composite price or earnings momentum measure.

5.1. Trading-based attention

The first measure of active investor attention is *Turnover*, the average daily share turnover of a stock over the past 12 months. Because active trading often requires deliberate decision-making, this measure likely captures active attention from all investors who have recently traded the stock. At the same time, because turnover reflects trading by both sophisticated and less sophisticated investors, it captures attention from a mixed investor clientele.

Table 7 reports the results. In Panel A, the differences in composite price momentum profits between the high- and low-*Turnover* quintiles are large and significant (135 bps per month in excess returns and 80–102 bps per month in factor alphas). These spreads are larger in magnitude compared to the spreads associated with the *Composite Attention* measure (108 bps in excess returns, and 64–91 bps in factor alphas). This finding, that price momentum is stronger among stocks with higher trading volume, is also consistent with Lee and Swaminathan (2000).

Conversely, in Panel B, the differences in composite earnings momentum profits between high- and low-*Turnover* quintiles are negative, –88 bps per month in excess returns and –100 to –119 bps per month in factor alphas, all highly significant. These are comparable to the corresponding spreads based on the *Composite Attention* measure (–87 bps in excess returns and –98 to –113 bps in factor alphas). Panel C shows that the difference in price and earnings momentum profits for the High–Low attention spread is strongly significant, indicating that greater *Turnover* both enhances price momentum and dampens earnings momentum, consistent with our hypothesis.

The similarity between the results for *Turnover* and *Composite Attention* is consistent

with the fact that both measures capture attention from both sophisticated and less sophisticated investors. In this interpretation, attention from sophisticated investors facilitates the timely incorporation of earnings news, reducing underreaction and weakening earnings momentum, whereas attention from less sophisticated investors amplifies extrapolative overreaction, strengthening price momentum.

5.2. Active attention by sophisticated investors

The next two active attention proxies are measures of sophisticated investors' information acquisition activities via the actual searching and downloading of information about a stock.

Bloomberg AIA is based on Bloomberg readership activities associated with a stock and, given that most of the Bloomberg subscribers are institutions, this measure reflects the attention of more sophisticated institutional investors (Ben-Rephael et al. 2017). Table 8 reports the two-way sorted portfolio analysis for this measure. Panel A indicates that price momentum profits tend to be somewhat larger for the high-*Bloomberg AIA* quintile than for the low-*Bloomberg AIA* quintile, but the differences are not statistically significant.

Regarding earnings momentum, Panel B finds that the profits are always substantially larger for the low-*Bloomberg AIA* quintile than they are for the high-*Bloomberg AIA* quintile. The High–Low attention differences in profits range from 132 to 159 bps per month and are all highly significant.[15]

These results are consistent with the finding of Ben-Rephael et al. (2017) that increased institutional attention helps reduce underreaction to earnings news. At the same time, the absence of statistically significant differences in price momentum profits across *Bloomberg AIA* quintiles suggests that institutional investors are less likely to contribute to overreaction-based price momentum.

We next examine *Edgar Downloads*, which tracks downloads of SEC filings from the

[15]Note that the earnings momentum profits for the high-*Bloomberg AIA* quintile are significantly negative for the q-factor model alphas and the DHS alpha. This likely reflects these models' success in explaining earnings momentum. Overcontrol may also come from the orthogonalization procedure itself. When we re-estimate Panel B using the unorthogonalized earnings momentum measure, all profit measures for the high-*Bloomberg AIA* quintile are statistically insignificant. Importantly, the High–Low portfolio continues to have significantly negative earnings momentum profits across all return measures, supporting the attention hypothesis.

EDGAR database. Such downloads likely reflect attention from both institutional investors and relatively sophisticated retail investors who pay attention to firm fundamentals. Table 9 reports that price momentum profits are substantially higher for firms in the high-*Edgar Downloads* quintile compared to firms in the low-*Edgar Downloads* quintile, by 92–137 bps per month. Conversely, earnings momentum profits are substantially lower for firms with the high-*Edgar Downloads* quintile compared to firms in the low-*Edgar Downloads* quintile, by 129–145 bps per month. Panel C shows that the difference in the directional predictions of attention on price versus earnings momentum is strongly significant. Thus, the attention measure based on *Edgar Downloads* is consistent with a dual role of attention, in which it both attenuates underreaction to earnings news and amplifies price overreaction.

The results also suggest that *Edgar Downloads* may serve multiple roles. Some downloads likely reflect rational information acquisition and processing that help investors better digest and respond to earnings news, while others may be associated with investor biases. The view that EDGAR users are not uniformly rational is also supported by (Liaukonytė and Žaldokas, 2022), who find that television advertising can trigger increases in EDGAR queries and is linked to greater total trading volume as well as increased trading activity on Robinhood, a popular retail trading platform.

5.3. Active attention by less sophisticated investors

Next, we consider active attention measures that correspond to less sophisticated investors, including those who participate in investment-related social media and those who actively search for stock ticker symbols online.

We begin with attention measures constructed with social media data that have attracted increasing interest in the literature.[16] The first measure is *Social Attention*, which extracts the first principal component of attention across three social media platforms—StockTwits, X, and Seeking Alpha. Table 10, Panel A, shows large price momentum profits for the high-attention quintile, ranging from 162 to 309 bps per month, whereas profits are insignificant for the low-attention quintile. The High–Low attention profit differential is substantial,

[16]See Cookson, Mullins, and Niessner (2024b); Hwang (2026) for surveys of the emerging financial social media literature.

ranging from 234 to 308 bps per month, suggesting that social media attention substantially amplifies the overreaction-based price momentum.

For earnings momentum, Panel B finds that the strongest earnings momentum profit is from the low-attention quintile, ranging from 146 to 200 bps per month, whereas profits are insignificant for high-attention quintiles. The earnings momentum spreads between the high- and low-*Social Attention* quintiles are also large, ranging from 180–265 bps per month. Panel C shows that the differential predictions of attention for price versus earnings momentum are all economically large and statistically significant.

We obtain qualitatively similar results when examining individual social media platforms. For brevity, these results are reported in the Internet Appendix. Using *StockTwits* (Table IA.4) and *Tweets* (Table IA.5) as attention measures, we find that higher social media attention is generally associated with stronger price momentum and weaker earnings momentum. Together, these findings indicate that social media attention amplifies overreaction-based price momentum while attenuating underreaction-based earnings momentum.

The fact that social media attention is strongly linked to overreaction patterns in prices supports the view that social interaction is an important channel through which behavioral biases are transmitted into asset prices (Hirshleifer, 2020). Consistent with this view, Cookson et al. (2024a) document a negative association between social media attention and future stock returns, suggesting that social media attention is associated with overvaluation and investor behavioral biases. At the same time, our finding that social media attention attenuates earnings momentum suggests that such attention, when used for information processing, can contribute to price discovery. In line with this interpretation, Chen et al. (2014) and Cookson et al. (2024a) provide evidence that sentiment extracted from social media posts contain valuable information about future returns. Our findings further this line of research and suggest that social media attention can be crucial in understanding different asset pricing anomalies.

The next measure is *Google ASV*, which captures the intensity of Google searches for a stock's ticker, and is commonly interpreted as reflecting the attention of less sophisticated retail investors (Da et al. 2011). Table IA.2 reports the two-way sorted portfolio analysis results. In Panel A, price momentum profits are largely absent across all attention quintiles,

and the price momentum spread between the high- and low-attention quintiles is also statistically insignificant. In Panel B, earnings momentum profits are positive and statistically significant in the low-attention quintile, and the profits are much smaller and statistically insignificant in the high-attention quintile. While this pattern is consistent with the hypothesis that attention reduces earnings momentum, the difference between the high- and low-attention quintiles is not statistically significant.

The lack of statistical significance when interacting *Google ASV* with momentum anomalies may be attributable to the fact that our analyses focus on a longer horizon, following the setup established in the momentum anomalies literature. All of our attention proxies are measured over the past 12 months, price momentum is measured over the past six months to one year, and earnings momentum is measured using earnings announcements or analyst forecast revisions from the past six months. As a result, higher-frequency variation in Google search activity that may be important for short-horizon return and trading dynamics is likely averaged out in our setting.[17]

We explore the possibility that a higher-frequency measure of Google-based attention interacts more strongly with the momentum anomalies. Specifically, we use the most recent month of Google abnormal search volume instead of the average over the past 12 months, and we restrict the sample to the one examined by Da et al. (2011), covering the period 2004–2008 and excluding stocks with prices less than five dollars. We further exclude stocks in the smallest size quintile (based on NYSE breakpoints) and momentum-crash months (proxied by down-market states). Panel A of Table IA.3 in the Internet Appendix provides evidence of stronger price momentum profits associated with higher recent Google attention. However, when we expand the sample through 2021, the difference in price momentum profits between the high- and low-attention quintiles, while still positive, is no longer significant

[17]Previous studies have found strong evidence that such higher-frequency Google search-based attention measure is directly associated with future returns. For example, Da et al. (2011) find that stocks that experience a weekly increase in Google search volume, relative to the variable's past eight-week median, earn higher abnormal returns over the subsequent two weeks, followed by an eventual reversal. This evidence is consistent with Barber and Odean (2008), who show that, since individual investors are net buyers of attention-grabbing stocks, an increase in individual investor attention leads to temporary overvaluation of the stock. Similarly, Da et al. (2026) provide evidence that daily aggregate retail attention negatively predicts market returns and is linked to aggregate retail order imbalance at a one-week horizon. While these papers examine Google search and returns, they do not investigate directly whether the Google measure moderates or exacerbates momentum effects.

(Panel B). Hence, the relationship between Google-based attention and price momentum remains inconclusive even for the higher frequency measure.

5.4. Passive attention

We next consider passive attention measures, where investors are exposed to information or distraction events without active seeking.

The first passive proxy is *News Articles*, defined as the number of news articles and press releases about a firm reported in RavenPack. This measure captures the supply of publicly available information rather than investors' deliberate attention. Table 11 reports the results. Panel A shows no significant difference in price momentum profits between high- and low-*News Articles* quintiles. In contrast, Panel B shows that earnings momentum profits are strong and highly significant among stocks with low news coverage, but are insignificant among stocks with high news coverage. These findings are consistent with greater news coverage facilitating faster and more accurate incorporation of earnings information, thereby reducing earnings underreaction, while not contributing to overreaction-based price momentum.

The second proxy is *–Distraction*, which captures variation in shareholder attention when their holdings in unrelated sectors experience extreme returns. Unlike other attention measures that directly proxy for attention to a given stock, this distraction-based measure relies on shocks originating in unrelated sectors, making it the least direct proxy for stock-level attention. As shown in Table 1, this measure differs markedly from other attention proxies. Whereas most attention measures are positively correlated with one another, *–Distraction* is negatively correlated with seven of the eight other measures, with correlations ranging from -0.03 to -0.11. This weak and negative correlation suggests that periods of heightened direct attention to a stock rarely coincide with periods when its key institutional investors are distracted by shocks elsewhere in their portfolios. As a result, findings based on this distraction-based proxy might not be directly comparable to those based on the other attention measures.

Table IA.6 presents the results. Panel A shows that *–Distraction* is negatively associated with price momentum profits, a pattern that contrasts with the results from other attention

measures. This pattern likely reflects the construction of the measure, which places greater weight on shareholders with large ownership stakes and for whom the firm represents a large share of their portfolio. As a result, the measure might be more sensitive to momentum crash periods, during which highly exposed investors are more likely to rapidly unwind positions to reduce risk. Consistent with this interpretation, Panel C of Table IA.8 shows that once down-market state months are excluded, the association between price momentum and *–Distraction* becomes largely insignificant. Similarly, Panel B of Table IA.6 shows no meaningful differences in earnings momentum profits across attention quintiles. Overall, this proxy does not exhibit a robust relationship with either price or earnings momentum. While *–Distraction* may capture institutional shareholders' monitoring of corporate decisions (Kempf et al. 2017), it does not appear to be systematically related to momentum anomaly profits.

6. Further analyses

This section provides further analyses using cross-sectional regressions and independently sorted portfolios, takes a closer look at price momentum profits and reversals, and considers the effect of firm size.

6.1. Cross-sectional regressions and independent portfolio sorts

To examine the robustness of our results, we estimate Fama and MacBeth (1973) cross-sectional regressions in Table 12. This analysis can control for additional stock characteristics and any formation period differences in the momentum spread for high versus low attention stocks. It can also serve to test the attention hypothesis' separate predictions for price and earnings momentum in one single estimation.

Each month, we regress the month t return of stocks on the following explanatory variables observed at month $t-1$. *Composite Attention Quintile* is a stock's quintile rank based on its *Composite Attention*. *Composite Price Momentum Score*, scaled between 0 and 1, is defined as a stock's average percentile rank based on *Past 6-mth Ret* and *Past 11-mth Ret*. *Composite Earnings Momentum Score*, also ranging between 0 and 1, is defined as a

stock's average percentile rank based on *SUE*, *ABR*, and *FREV*. Our key variables of interest are the two interaction terms: *Composite Price Momentum Score*×*Composite Attention Quintile* and *Composite Earnings Momentum Score*×*Composite Attention Quintile*. The control variables include beta, log firm size, log book-to-market ratio, the lagged return, and idiosyncratic volatility.[18]

The dual-role hypothesis predicts a positive coefficient on the former interaction and a negative coefficient on the latter. Table 12 presents the results. Column (1) reports the baseline specification including price momentum and controls, while column (2) reports the baseline specification including earnings momentum and controls. In both cases, the momentum coefficients are positive and statistically significant, consistent with the profitability of these strategies.

Column (3) introduces the interaction between price momentum and attention and shows a positive and significant coefficient, while the standalone price momentum term becomes insignificant. This finding is consistent with the hypothesis that attention amplifies the mispricing underlying price momentum. Column (4) introduces the interaction between earnings momentum and attention and yields a negative and significant coefficient, consistent with the view that attention mitigates underreaction to earnings news and improves price discovery. Column (5) includes both momentum measures and both interaction terms simultaneously, and the coefficients on the interaction terms remain robust. Overall, the regression results corroborate the portfolio-based evidence and confirm the dual role of investor attention in strengthening price momentum while dampening earnings momentum.

As an additional robustness check, we replace the sequentially sorted portfolio analysis with independently sorted portfolios for our main results. Table IA.7 in the Internet Appendix shows that the results remain very similar to those in Table 6, both in magnitude and statistical significance.

[18]Beta is estimated by regressing the last three years of the stock's monthly returns on the returns of the CRSP value-weighted market index, the book-to-market ratio is computed following Fama and French (2006), lagged return is the stock's return in month $t-1$, and idiosyncratic volatility is the standard deviation of daily residuals of the stock's daily returns regressed against the daily returns of the Fama and French (1993) three factors in month $t-1$.

6.2. A closer look at price momentum

This subsection takes a closer look at the profits from price momentum strategies. For example, Table 6, column (1), presents the returns in excess of the risk-free rate from the long–short portfolios based on price momentum. The results show that the profits for the high-attention quintile are 91 bps and highly significant (t-statistics 4.92), whereas the profits for the low-attention quintile are -16 bps and statistically insignificant (t-statistics 0.85). The differential between the high- and low-attention quintiles is highly significant, at 108 bps. This pattern is consistent with the hypothesis that investor attention strengthens price momentum.

Table 6, columns (2)–(5) present the corresponding abnormal returns, measured as the q-4 alpha, the q-5 alpha, the FF6 alpha, and the DHS alpha. The high- and low-attention price momentum profit differentials remain robust, ranging from 64 to 91 bps and statistically significant, consistent with the core prediction of our hypothesis. However, once returns are adjusted using these factor models, price momentum profits are no longer positive and statistically significant for high-attention stocks, while the profits for low-attention stocks become negative and statistically significant. This pattern indicates a substantial downward shift in price momentum profitability across all attention deciles when returns are risk-adjusted using recent factor models, compared to subtracting the risk-free rate. A similar pattern appears in other price momentum-related tables.

Our analysis suggests two possible explanations for this pattern: the increasing ability of modern factor models to explain price momentum profits, and the impact of momentum crashes during the sample period.

The first explanation relates to the success of recently developed factor models in explaining price momentum profits. Because price momentum is a well-known anomaly, more recently proposed factor models seeking adoption in the asset-pricing literature are likely to capture momentum returns to varying degrees. In fact, the FF6 model explicitly includes a price momentum factor. It is therefore not surprising that price momentum profits become substantially smaller, or even turn negative, once abnormal returns are adjusted using these factors.

To see this, we compare the results with abnormal returns computed with an older factor model, i.e., the Fama–French three-factor model (FF3), which is expected to be less effective in explaining price momentum. Panel A of Table IA.8 in the Internet Appendix replicates Table 6 and presents the two-way sorted portfolio results, with the FF3 alphas displayed in the last column. The results show that abnormal returns to price momentum portfolios are close to zero for low-attention stocks but are highly positive and statistically significant for high-attention stocks, with the differential entirely driven by the high-attention group. This pattern closely mirrors that observed for excess returns in column 1 of Table 6.

Crucially, across all the factor models used in Tables 6 and IA.8, the differences in price momentum returns between the high- and low-attention stocks remain positive and highly significant. This suggests that while newer factor models are successful in removing average price momentum profits, they are unable to eliminate the differential profitability across attention groups. This finding underscores the role of investor attention in shaping cross-sectional differences in price momentum profitability.

The second reason for the across-the-board downward shift in price momentum profits is due to the presence of momentum crashes in our sample period. Daniel et al. (1998), Cooper et al. (2004) and Wang and Xu (2015) find that the price momentum strategy sometimes suffers crashes and that these crashes are often preceded by down-market states. For example, a very prominent crash is the year 2009, where the UMD factor's return for the year was −83.8%. Motivated by this evidence, we use down-market states as an ex ante indicator of momentum crash risk and exclude months in which the buy-and-hold market return over the prior 36-month period is negative. Panel B of Table IA.8 in the Internet Appendix reports the results, showing that the negative alphas for the low-attention quintile are substantially attenuated, while the alphas for the high-attention quintile are uniformly positive and statistically significant in all but one case. Crucially, our key finding of the positive relationship between attention and price momentum remains robust.[19]

[19] A related question is whether earnings momentum also experiences occasional crashes, in comparison with price momentum. Figure IA.1 presents a histogram of the PEAD factor versus the UMD factor. The UMD factor exhibits substantially stronger negative skewness than the PEAD factor. This contrast suggests that earnings momentum is much less exposed to crash risk, and that momentum crashes are likely not a major concern for earnings-based momentum strategies.

6.3. Reversals

We now examine the implications for longer-term returns. If high investor attention drives overreaction to past price movements, we hypothesize that high-attention stocks will exhibit stronger long-term reversals following the initial momentum period. In contrast, as earnings momentum is primarily attributed to underreaction, we do not anticipate significant long-term reversals for this strategy.

Figure 1 plots the cumulative monthly q-4 alphas for long–short portfolios formed on *Composite Price Momentum (Orthogonalized)*, comparing high- and low-attention stocks. Following Hong et al. (2000), we exclude microcap stocks (the smallest size quintile based on NYSE breakpoints), as their long-run performance is often noisy. We also omit momentum crash months, proxied by down-market states.

The figure shows that price momentum profits are substantially stronger for high-attention stocks than for low-attention stocks during months 1 to 7. Importantly, these high-attention portfolios also experience more pronounced reversals in the subsequent period, from months 8 to 24. Table IA.9 in the Internet Appendix details the month-by-month alphas. The reversals commence in month 8, with the return differential between high- and low-attention quintiles turning negative for all subsequent months and being statistically significant in six of them. This evidence supports our hypothesis that high attention is associated with greater short-term overreaction and, consequently, stronger long-run reversals in price momentum.

Panel A of Figure 2 presents the cumulative monthly q-4 alphas for long–short portfolios based on *Composite Earnings Momentum (Orthogonalized)*. It shows that low-attention stocks generate higher earnings momentum profits that do not reverse in the 24-month holding period. In contrast, there are little earnings momentum profits for high-attention stocks. These results reinforce the view that orthogonalized earnings momentum captures an underreaction to earnings news, and that high investor attention mitigates this underreaction.

Finally, we contrast the long-run performance of the orthogonalized earnings momentum strategy with that of the unorthogonalized *Composite Earnings Momentum* strategy. Panel B of Figure 2 displays the results for the unorthogonalized strategy, showing that reversals emerge after around month nine (see also Table IA.10 of the Internet Appendix). This

pattern is consistent with Bathke et al. (2019). The absence of such reversals in the orthogonalized strategy suggests that these reversals likely stem from the component of earnings momentum that overlaps with price momentum, which is known to exhibit long-run reversals. By removing this shared element, the orthogonalization procedure effectively isolates the unique component of each anomaly, allowing a clearer analysis of the distinct mechanisms driving price and earnings momentum.

6.4. The effect of firm size

A potential concern is that several of our attention proxies are highly correlated with firm size, as larger firms naturally command more media coverage and investor interest. To ensure our results are not merely capturing a size effect, we isolate the impact of attention by constructing a size-controlled composite measure. To do this, we first orthogonalize each of the nine individual attention proxies with respect to firm size. For each month, we perform cross-sectional regressions of the attention measure on the firm's market capitalization from the previous month. The residuals from these regressions, which capture the variation in attention not explained by size, are then used to sort stocks into quintiles. Finally, we aggregate these quintile ranks to create a new *Composite Attention (Size-Controlled)* measure.

Table IA.11 presents the results from interacting this size-controlled attention measure with our orthogonalized price and earnings momentum measures. Panel A reports that the difference in price momentum profits between the high- and low-attention quintiles ranges from 69–111 bps per month. This spread is economically significant and comparable to the 64–108 bps per month found using the original attention measure (as shown in Table 6).

For earnings momentum, Panel B shows that profits remain substantially lower for high-attention stocks, with spreads between the high- and low-attention quintiles ranging from –52 to –75 bps per month. While the magnitude of this effect is smaller than the –87 to –113 bps observed without the size control (Table 6), the results remain highly statistically significant.

Therefore, we conclude that the role of investor attention on price and earnings momentum is a distinct phenomenon and not simply a proxy for the effect of firm size.

7. Conclusion

In this paper, we examine the hypothesis that investor attention plays a dual role in determining the stock market's reaction to information. Specifically, we posit that attention alleviates stock market underreaction to value-relevant information, such as earnings news, while attention can also amplify behavioral biases that drive the overreaction-based price momentum.

Using a comprehensive set of both established and recently proposed attention proxies, we analyze their relationship with multiple price and earnings momentum anomalies. We find that firms attracting greater investor attention exhibit stronger price momentum but weaker earnings momentum. These results support the dual role hypothesis: attention enhances price momentum but weakens earnings momentum.

We further document significant heterogeneity across attention measures, driven by the distinction between active and passive attention and investor sophistication. Institutional attention, measured by Bloomberg readership, reduces earnings momentum but has no significant link to price momentum. This supports the view that active institutional attention facilitates rational information processing and price discovery. A similar beneficial effect is observed for news supply-based attention. Conversely, retail-oriented and social media-based attention strengthens price momentum, consistent with the notion that active attention from less sophisticated investors amplifies behavioral biases. Interestingly, this retail attention also dampens earnings momentum, suggesting that social media participants are not uniformly unsophisticated, or that even less experienced investors contribute to price efficiency when focusing on fundamental signals.

Overall, our results suggest that investor attention plays distinct and critical roles in underreaction- and overreaction-based anomalies, highlighting that the interaction between investor type and information processing is central to understanding market efficiency.

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Table 1: Descriptive Statistics and Correlation of Attention Measures

This table reports the descriptive statistics (Panel A) and correlation matrix (Panel B) of the stock-level attention measures. *Turnover* is the stock's monthly average daily turnover in the last 12 months. *Bloomberg AIA* is the monthly average Bloomberg Daily Maximum Readership (DMR) computed according to Ben-Rephael et al. (2017), averaged over the last 12 months. *Edgar Downloads* is the last 12-month average number of monthly non-robot searches in the SEC's EDGAR system, using data provided by James Ryans computed using the Loughran and McDonald (2017) method. *Social Attention* is the first principal component of social media messages/posts on StockTwits, X, and Seeking Alpha, with data provided by Cookson et al. (2024a) where the daily values are averaged within the month before computing the past 12-month average as *Social Attention*. *StockTwits* is the average monthly number of messages sent by investors on the StockTwits platform in the last 12 months. *Tweets* is the average monthly number of tweets and retweets sent by investors on the X (formerly Twitter) platform in the last 12 months. For Google search, we compute the percentage change between Google's daily Search Volume Index (SVI) for a stock and its past six-month median (Da et al., 2011). This value is averaged for each month and the past 12-month average is defined as *Google ASV*. *News Articles* is the last 12-month average monthly number of news articles reported by RavenPack. *–Distraction* is −1 multiplied by the quarterly institutional distraction measure computed following Kempf et al. (2017), averaged over the last 12 months. The reported mean, stdev, firms per month, and percentiles in Panel A are time-series averages of the relevant cross-sectional statistics. Panel B shows the time-series average of the cross-sectional correlation coefficients of the quintile ranks of the attention measures. Ranks from 1 to 5 are assigned by sorting stocks into quintiles each month according to each measure. *Composite Attention* is defined as the monthly average quintile ranks of the individual attention measures when available.

Panel A: Descriptive Statistics

Attention Measure	Mean	Stdev	Sample	#Firms per mth	Percentiles				
					1%	20%	Median	80%	99%
Turnover	0.005	0.0096	196701–202212	3,745	0.0002	0.0010	0.0030	0.0060	0.0300
Bloomberg AIA	0.086	0.094	201003–202112	2,276	0.000	0.013	0.064	0.135	0.430
Edgar Downloads	446.7	890.5	200302–201712	2,987	29.0	142.6	263.3	513.8	3780.4
Social Attention	-0.029	0.488	201202–202212	709	-0.246	-0.171	-0.097	0.014	1.170
StockTwits	130.2	980.9	200904–202212	3,070	2.0	8.4	17.8	63.8	2056.6
Tweets	309.4	1536.7	201301–201912	3,364	5.9	46.2	102.9	265.0	3490.0
Google ASV	0.040	0.090	200408–202112	1,489	-0.212	-0.011	0.034	0.090	0.300
News Articles	90.2	502.9	200102–202210	3,620	4.4	18.3	33.9	70.0	1019.1
–Distraction	-0.157	0.035	198001–202212	4,231	-0.230	-0.179	-0.166	-0.130	-0.070

Panel B: Correlation Matrix

Attention Measure	Correlation between Quintile Ranks								
	Turnover	Bloomberg	Edgar	Social Attention	Stock-Twits	Tweets	Google	News	–Distraction
Turnover	1								
Bloomberg AIA	0.368	1							
Edgar Downloads	0.580	0.560	1						
Social Attention	0.123	0.411	0.439	1					
StockTwits	0.651	0.494	0.568	0.286	1				
Tweets	0.668	0.573	0.676	0.463	0.766	1			
Google ASV	0.053	0.089	0.055	0.068	0.119	0.128	1		
News Articles	0.557	0.578	0.723	0.475	0.541	0.685	0.068	1	
–Distraction	-0.035	-0.078	-0.113	-0.107	-0.031	-0.101	0.020	-0.093	1
Composite Attention	0.807	0.714	0.798	0.649	0.792	0.849	0.374	0.782	0.484

Table 2: Correlations between Price and Earnings Momentum Measures

This table reports the average monthly correlations of the quintile ranks of the momentum measures. Ranks from 1 to 5 are assigned by sorting all stocks into quintiles every month according to each measure. The correlations are computed for each monthly cross-section and the time-series averages are reported in Panel A for price momentum measures and in Panel B for earnings momentum measures. Price momentum measures are defined as follows: *Past 6-mth Ret* is the firm's buy-and-hold return from month $t-7$ to month $t-2$; and *Past 11-mth Ret* is the firm's buy-and-hold return from month $t-12$ to month $t-2$. Earnings momentum measures are defined as follows: *SUE*, the standardized unexpected earnings, is the firm's quarterly earnings (Compustat item IBQ) less its earnings over the same quarterly horizon in the prior year, with this difference normalized by the standard deviation of the firm's previous eight changes in earnings; *ABR* is the cumulative market-adjusted return over the [-2,+1] trading days around the most recent quarterly earnings announcement date (item RDQ); and *FREV* is the six-month moving average of past monthly changes in analysts' FY1 consensus earnings forecast (I/B/E/S item meanest) scaled by the prior month stock price. Composite momentum measures are defined as follows: The composite price momentum measure is computed by averaging the quintile ranks of *Past 6-mth Ret* and *Past 11-mth Ret* every month when available; and the composite earnings momentum measure is computed by averaging the quintile ranks of *SUE*, *ABR*, and *FREV* every month when available.

Panel A: Correlations of the Price Momentum Measures

Measure	Sample	#Firms per mth	Mean	Correlation between Quintile Ranks		
				Past 6-mth ret	Past 11-mth ret	Composite Price Momentum
Past 6-mth Ret	196701–202212	4,011	0.072	1		
Past 11-mth Ret	196701–202212	4,013	0.138	0.682	1	
Composite Price Momentum	196701–202212	4,013	3.000	0.917	0.917	1
Composite Earnings Momentum	197107–202212	3,774	2.991	0.309	0.331	0.349

Panel B: Correlations of the Earnings Momentum Measures

Measure	Sample	#Firms per mth	Mean	Correlation between Quintile Ranks		
				SUE	ABR	FREV
SUE	197109–202212	3,648	0.180	1		
ABR	197107–202212	3,702	0.003	0.198	1	
FREV	197609–202212	2,712	0.044	0.343	0.115	1
Composite Earnings Momentum	197107–202212	3,774	2.991	0.753	0.698	0.719
Composite Price Momentum	196701–202212	4,013	3.000	0.323	0.172	0.318

Table 3: Composite Attention and Price Momentum

We conduct a two-way sorted portfolio analysis and report the relation between *Composite Attention* and long–short price momentum portfolio returns. Each month, we sort stocks into quintiles based on *Composite Attention*, and then within each attention quintile we double-sort stocks into quintiles according to their price momentum measure. Stocks are held for one month and the portfolio returns are equal-weighted. The average long–short price momentum portfolio excess returns ($R_p - R_f$) and alphas from the q-4, q-5, Fama and French (2018) six-factor (FF6), and the DHS behavioral factor models are reported for each attention quintile, with the spread between the high- and low-attention quintiles reported as High–Low. Panels A and B report the results with *Past 6-mth Ret* and *Past 11-mth Ret*, respectively as the price momentum measure. See Tables 1 and 2 for variable descriptions. *, **, and *** denote statistical significance of the percentage abnormal monthly returns at the 10%, 5%, and 1% levels, respectively, with the associated t-statistics in parentheses.

Composite Attention Quintile	Price Momentum Long–Short Portfolio Returns				
	$R_p - R_f$	q-4 alpha	q-5 alpha	FF6 alpha	DHS alpha
	Panel A: Past 6-mth Ret				
Low	0.123	-0.353**	-0.394**	-0.362**	-0.429**
	(0.65)	(-1.97)	(-2.04)	(-2.18)	(-2.05)
2	0.593***	0.039	0.012	0.003	-0.056
	(2.98)	(0.21)	(0.06)	(0.02)	(-0.26)
3	0.747***	0.127	0.052	0.047	-0.117
	(3.43)	(0.62)	(0.24)	(0.28)	(-0.50)
4	0.930***	0.148	-0.032	0.052	-0.130
	(3.52)	(0.61)	(-0.13)	(0.26)	(-0.46)
High	1.171***	0.319	0.097	0.156	0.008
	(3.84)	(1.10)	(0.31)	(0.73)	(0.02)
High–Low	1.048***	0.672***	0.492*	0.518**	0.437
	(4.35)	(2.69)	(1.83)	(2.42)	(1.62)
	Panel B: Past 11-mth Ret				
Low	0.495**	-0.008	-0.101	-0.060	-0.082
	(2.52)	(-0.04)	(-0.51)	(-0.37)	(-0.37)
2	0.884***	0.237	0.130	0.209	0.231
	(4.28)	(1.25)	(0.64)	(1.34)	(1.02)
3	0.957***	0.244	0.110	0.152	-0.007
	(4.08)	(1.12)	(0.47)	(0.93)	(-0.03)
4	1.127***	0.226	-0.016	0.146	0.046
	(4.16)	(0.95)	(-0.06)	(0.79)	(0.16)
High	1.432***	0.449	0.191	0.310*	0.280
	(4.57)	(1.57)	(0.62)	(1.65)	(0.82)
High–Low	0.936***	0.457*	0.292	0.370*	0.361
	(3.96)	(1.89)	(1.13)	(1.84)	(1.34)

Table 4: Composite Attention and Earnings Momentum

We conduct a two-way sorted portfolio analysis and report the relation between *Composite Attention* and long–short earnings momentum portfolio returns. Each month, we sort stocks into quintiles based on *Composite Attention*, and then within each attention quintile we double-sort stocks into quintiles according to their earnings momentum measure. Stocks are held for one month and the portfolio returns are equal-weighted. The average long–short earnings momentum portfolio excess returns ($R_p - R_f$) and alphas from the q-4, q-5, Fama and French (2018) six-factor (FF6), and the DHS behavioral factor models are reported for each attention quintile, with the spread between the high- and low-attention quintiles reported as High–Low. Panels A, B, and C report the results with *SUE*, *ABR*, and *FREV*, respectively as the earnings momentum measure. See Tables 1 and 2 for variable descriptions. *, **, and *** denote statistical significance of the percentage abnormal monthly returns at the 10%, 5%, and 1% levels, respectively, with the associated t-statistics in parentheses.

Composite Attention Quintile	Earnings Momentum Long–Short Portfolio Returns				
	$R_p - R_f$	q-4 alpha	q-5 alpha	FF6 alpha	DHS alpha
		Panel A: SUE			
Low	1.753***	1.441***	1.462***	1.511***	1.455***
	(16.26)	(14.37)	(13.52)	(14.69)	(12.55)
2	1.330***	1.013***	1.057***	1.102***	1.033***
	(11.91)	(10.00)	(9.68)	(10.65)	(8.52)
3	1.318***	0.947***	1.052***	1.069***	1.016***
	(10.69)	(9.09)	(9.41)	(9.80)	(7.56)
4	1.244***	0.706***	0.749***	0.869***	0.826***
	(7.84)	(5.58)	(5.49)	(6.49)	(4.86)
High	1.180***	0.520***	0.456***	0.705***	0.557***
	(6.76)	(3.81)	(3.10)	(4.89)	(3.09)
High–Low	-0.574***	-0.921***	-1.006***	-0.805***	-0.899***
	(-3.78)	(-6.34)	(-6.43)	(-5.54)	(-5.56)
		Panel B: ABR			
Low	1.203***	1.127***	1.143***	1.112***	1.089***
	(12.60)	(11.07)	(10.41)	(11.12)	(10.32)
2	1.012***	0.866***	0.825***	0.863***	0.752***
	(9.87)	(8.03)	(7.09)	(8.18)	(7.02)
3	1.029***	0.941***	0.804***	0.914***	0.691***
	(9.98)	(8.81)	(7.04)	(8.82)	(6.69)
4	0.964***	0.761***	0.684***	0.775***	0.512***
	(8.46)	(6.57)	(5.49)	(6.86)	(4.52)
High	0.884***	0.668***	0.490***	0.685***	0.299**
	(6.85)	(5.26)	(3.61)	(5.49)	(2.46)
High–Low	-0.318**	-0.458***	-0.653***	-0.427***	-0.790***
	(-2.16)	(-3.01)	(-4.01)	(-2.80)	(-5.19)
		Panel C: FREV			
Low	1.131***	0.919***	0.879***	1.000***	1.064***
	(9.02)	(7.06)	(6.34)	(7.66)	(7.74)
2	0.771***	0.554***	0.585***	0.687***	0.707***
	(6.29)	(4.43)	(4.39)	(5.49)	(5.26)
3	0.848***	0.569***	0.539***	0.632***	0.694***
	(6.20)	(4.20)	(3.73)	(4.77)	(4.65)
4	0.664***	0.361**	0.448***	0.495***	0.490***
	(4.44)	(2.45)	(2.86)	(3.47)	(3.06)
High	0.699***	0.238	0.327*	0.325*	0.334*
	(3.70)	(1.32)	(1.70)	(1.95)	(1.69)
High–Low	-0.432**	-0.681***	-0.552***	-0.676***	-0.730***
	(-2.20)	(-3.40)	(-2.59)	(-3.58)	(-3.56)

Table 5: Composite Attention and Composite Price and Earnings Momentum

We conduct a two-way sorted portfolio analysis and report the relation between *Composite Attention* and long–short portfolio returns for composite momentum portfolios. Each month, we sort stocks into quintiles based on *Composite Attention*, and then within each attention quintile we double-sort stocks into quintiles according to their *Composite Price Momentum* or *Composite Earnings Momentum* measures. Stocks are held for one month and the portfolio returns are equal-weighted. The average long–short momentum portfolio excess returns ($R_p - R_f$) and alphas from the q-4, q-5, Fama and French (2018) six-factor (FF6), and the DHS behavioral factor models are reported for each attention quintile, with the spread between the high- and low-attention quintiles reported as High–Low. Panels A and B report the results for the *Composite Price Momentum* measure and the *Composite Earnings Momentum* measure, respectively. Panel C reports the return differences between Panels A and B for the Low, High, and High–Low portfolios. See Tables 1 and 2 for variable descriptions. *, **, and *** denote statistical significance of the percentage abnormal monthly returns at the 10%, 5%, and 1% levels, respectively, with the associated t-statistics in parentheses.

Composite Attention Quintile	Long–Short Portfolio Returns				
	$R_p - R_f$	q-4 alpha	q-5 alpha	FF6 alpha	DHS alpha
	Panel A: Composite Price Momentum				
Low	0.363*	-0.144	-0.171	-0.190	-0.213
	(1.74)	(-0.75)	(-0.82)	(-1.07)	(-0.98)
2	0.836***	0.217	0.194	0.151	0.162
	(3.81)	(1.07)	(0.89)	(0.88)	(0.71)
3	0.900***	0.209	0.126	0.074	0.015
	(3.74)	(0.96)	(0.54)	(0.44)	(0.06)
4	0.966***	0.118	-0.128	-0.048	-0.121
	(3.31)	(0.46)	(-0.47)	(-0.24)	(-0.41)
High	1.477***	0.594**	0.341	0.382*	0.260
	(4.67)	(2.11)	(1.13)	(1.94)	(0.82)
High–Low	1.114***	0.738***	0.512*	0.572***	0.473*
	(4.57)	(2.97)	(1.92)	(2.67)	(1.83)
	Panel B: Composite Earnings Momentum				
Low	1.963***	1.702***	1.703***	1.744***	1.751***
	(19.78)	(17.30)	(16.05)	(17.69)	(16.06)
2	1.507***	1.235***	1.269***	1.303***	1.213***
	(13.48)	(11.36)	(10.83)	(11.98)	(10.00)
3	1.488***	1.173***	1.144***	1.250***	1.118***
	(12.17)	(10.65)	(9.64)	(11.28)	(8.63)
4	1.342***	0.847***	0.874***	0.963***	0.861***
	(9.00)	(6.51)	(6.24)	(7.36)	(5.54)
High	1.303***	0.708***	0.594***	0.822***	0.562***
	(6.86)	(4.46)	(3.48)	(5.37)	(3.01)
High–Low	-0.660***	-0.994***	-1.109***	-0.921***	-1.188***
	(-3.71)	(-5.92)	(-6.15)	(-5.68)	(-6.60)
	Panel C: Composite Price- minus Earnings Momentum				
Low	-1.600***	-1.845***	-1.874***	-1.933***	-1.964***
	(-8.80)	(-10.33)	(-9.73)	(-11.61)	(-10.26)
High	0.174	-0.114	-0.253	-0.440**	-0.303
	(0.81)	(-0.51)	(-1.06)	(-2.47)	(-1.30)
High–Low	1.774***	1.731***	1.621***	1.493***	1.661***
	(8.42)	(7.73)	(6.72)	(6.95)	(7.08)

Table 6: Composite Attention and Composite Price and Earnings Momentum (Orthogonalized)

We conduct a two-way sorted portfolio analysis and report the relation between *Composite Attention* and long–short portfolio returns for the orthogonalized composite momentum portfolios. Each month, we sort stocks into quintiles based on *Composite Attention*, and then within each attention quintile we double-sort stocks into quintiles according to their orthogonalized composite price or earnings momentum measures. Stocks are held for one month and the portfolio returns are equal-weighted. The average long–short orthogonalized momentum portfolio excess returns ($R_p - R_f$) and alphas from the q-4, q-5, Fama and French (2018) six-factor (FF6), and the DHS behavioral factor models are reported for each attention quintile, with the spread between the high- and low-attention quintiles reported as High–Low. Panels A and B report the results for the orthogonalized composite price momentum measure and the orthogonalized composite earnings momentum measure, respectively. Panel C reports the return differences between Panels A and B for the Low, High, and High–Low portfolios. See Tables 1 and 2 for variable descriptions. *, **, and *** denote statistical significance of the percentage abnormal monthly returns at the 10%, 5%, and 1% levels, respectively, with the associated t-statistics in parentheses.

Composite Attention Quintile	Long–Short Portfolio Returns				
	$R_p - R_f$	q-4 alpha	q-5 alpha	FF6 alpha	DHS alpha
Panel A: Composite Price Momentum (Orthogonalized)					
Low	-0.163	-0.636***	-0.679***	-0.683***	-0.683***
	(-0.85)	(-3.62)	(-3.59)	(-4.28)	(-3.42)
2	0.335*	-0.155	-0.169	-0.255	-0.232
	(1.65)	(-0.81)	(-0.82)	(-1.62)	(-1.10)
3	0.505**	-0.095	-0.164	-0.277*	-0.285
	(2.28)	(-0.46)	(-0.75)	(-1.83)	(-1.26)
4	0.534**	-0.134	-0.361	-0.337**	-0.356
	(2.12)	(-0.59)	(-1.49)	(-1.97)	(-1.41)
High	0.912***	0.268	0.095	-0.007	-0.045
	(3.28)	(1.03)	(0.34)	(-0.04)	(-0.16)
High–Low	1.076***	0.905***	0.774***	0.676***	0.638***
	(4.92)	(3.95)	(3.14)	(3.48)	(2.73)
Panel B: Composite Earnings Momentum (Orthogonalized)					
Low	1.829***	1.730***	1.763***	1.792***	1.807***
	(19.68)	(17.61)	(16.64)	(18.34)	(17.49)
2	1.378***	1.293***	1.346***	1.386***	1.271***
	(14.71)	(13.15)	(12.72)	(14.14)	(12.46)
3	1.286***	1.178***	1.160***	1.298***	1.221***
	(12.35)	(10.90)	(9.95)	(11.84)	(10.57)
4	1.136***	0.927***	1.020***	1.094***	1.034***
	(10.22)	(8.25)	(8.44)	(9.46)	(8.38)
High	0.961***	0.634***	0.642***	0.812***	0.678***
	(7.16)	(4.97)	(4.66)	(5.98)	(4.70)
High–Low	-0.868***	-1.096***	-1.121***	-0.980***	-1.129***
	(-5.63)	(-7.17)	(-6.80)	(-6.26)	(-6.84)
Panel C: Composite Price- minus Earnings Momentum (Orthogonalized)					
Low	-1.992***	-2.366***	-2.442***	-2.475***	-2.490***
	(-8.63)	(-10.58)	(-10.13)	(-12.08)	(-10.27)
High	-0.049	-0.366	-0.547*	-0.819***	-0.723**
	(-0.16)	(-1.19)	(-1.65)	(-3.49)	(-2.26)
High–Low	1.943***	2.001***	1.895***	1.656***	1.767***
	(6.96)	(6.76)	(5.94)	(5.97)	(5.75)

Table 7: Turnover and Price- and Earnings Momentum

We conduct a two-way sorted portfolio analysis and report the relation between an attention proxy, *Turnover*, and long–short portfolio returns for the orthogonalized composite momentum portfolios. Each month, we sort stocks into quintiles based on the attention measure, and then within each attention quintile we double-sort stocks into quintiles according to their orthogonalized composite price or earnings momentum measures. Stocks are held for one month and the portfolio returns are equal-weighted. The average long–short orthogonalized momentum portfolio excess returns ($R_p - R_f$) and alphas from the q-4, q-5, Fama and French (2018) six-factor (FF6), and the DHS behavioral factor models are reported for each attention quintile, with the spread between the high- and low-attention quintiles reported as High–Low. Panels A and B report the results for the orthogonalized composite price momentum measure and the orthogonalized composite earnings momentum measure, respectively. Panel C reports the return differences between Panels A and B for the Low, High, and High–Low portfolios. See Tables 1 and 2 for variable descriptions. *, **, and *** denote statistical significance of the percentage abnormal monthly returns at the 10%, 5%, and 1% levels, respectively, with the associated t-statistics in parentheses.

Turnover Quintile	Long–Short Portfolio Returns				
	$R_p - R_f$	q-4 alpha	q-5 alpha	FF6 alpha	DHS alpha
Panel A: Composite Price Momentum (Orthogonalized)					
Low	-0.264	-0.617***	-0.633***	-0.645***	-0.729***
	(-1.45)	(-3.62)	(-3.45)	(-4.02)	(-3.87)
2	0.108	-0.413**	-0.440**	-0.546***	-0.524**
	(0.53)	(-2.19)	(-2.16)	(-3.42)	(-2.48)
3	0.263	-0.348*	-0.553**	-0.554***	-0.563**
	(1.15)	(-1.67)	(-2.46)	(-3.57)	(-2.45)
4	0.832***	0.211	0.003	-0.007	-0.015
	(3.27)	(0.91)	(0.01)	(-0.04)	(-0.06)
High	1.084***	0.401	0.258	0.160	0.119
	(3.88)	(1.53)	(0.92)	(0.90)	(0.42)
High–Low	1.348***	1.018***	0.891***	0.804***	0.847***
	(5.70)	(4.14)	(3.36)	(3.89)	(3.30)
Panel B: Composite Earnings Momentum (Orthogonalized)					
Low	1.821***	1.751***	1.762***	1.792***	1.835***
	(18.95)	(17.22)	(16.07)	(17.80)	(17.22)
2	1.759***	1.714***	1.752***	1.799***	1.690***
	(17.35)	(15.83)	(15.01)	(16.77)	(15.24)
3	1.205***	1.052***	1.133***	1.188***	1.171***
	(12.04)	(10.16)	(10.18)	(11.32)	(10.52)
4	0.863***	0.579***	0.640***	0.767***	0.680***
	(7.77)	(5.53)	(5.68)	(6.85)	(5.50)
High	0.945***	0.626***	0.624***	0.794***	0.641***
	(6.88)	(4.80)	(4.44)	(5.72)	(4.35)
High–Low	-0.875***	-1.125***	-1.139***	-0.998***	-1.193***
	(-5.57)	(-7.38)	(-6.93)	(-6.31)	(-7.13)
Panel C: Composite Price- minus Earnings Momentum (Orthogonalized)					
Low	-2.085***	-2.367***	-2.395***	-2.437***	-2.564***
	(-9.10)	(-10.60)	(-9.94)	(-11.51)	(-10.70)
High	0.139	-0.225	-0.365	-0.634**	-0.523
	(0.46)	(-0.71)	(-1.07)	(-2.56)	(-1.60)
High–Low	2.224***	2.143***	2.030***	1.803***	2.041***
	(7.47)	(6.78)	(5.96)	(6.13)	(6.18)

Table 8: Bloomberg AIA and Price- and Earnings Momentum

We conduct a two-way sorted portfolio analysis and report the relation between an attention proxy, *Bloomberg AIA*, and long–short portfolio returns for the orthogonalized composite momentum portfolios. Each month, we sort stocks into quintiles based on the attention measure, and then within each attention quintile we double-sort stocks into quintiles according to their orthogonalized composite price or earnings momentum measures. Stocks are held for one month and the portfolio returns are equal-weighted. The average long–short orthogonalized momentum portfolio excess returns ($R_p - R_f$) and alphas from the q-4, q-5, Fama and French (2018) six-factor (FF6), and the DHS behavioral factor models are reported for each attention quintile, with the spread between the high- and low-attention quintiles reported as High–Low. Panels A and B report the results for the orthogonalized composite price momentum measure and the orthogonalized composite earnings momentum measure, respectively. Panel C reports the return differences between Panels A and B for the Low, High, and High–Low portfolios. See Tables 1 and 2 for variable descriptions. *, **, and *** denote statistical significance of the percentage abnormal monthly returns at the 10%, 5%, and 1% levels, respectively, with the associated t-statistics in parentheses.

Bloomberg AIA Quintile	Long–Short Portfolio Returns				
	$R_p - R_f$	q-4 alpha	q-5 alpha	FF6 alpha	DHS alpha
Panel A: Composite Price Momentum (Orthogonalized)					
Low	0.169	0.555	0.330	0.361	0.192
	(0.40)	(1.29)	(0.76)	(0.90)	(0.46)
2	0.107	-0.052	-0.127	-0.240	-0.156
	(0.30)	(-0.15)	(-0.35)	(-0.76)	(-0.43)
3	0.456	0.786**	0.607*	0.537*	0.461
	(1.27)	(2.31)	(1.78)	(1.93)	(1.41)
4	0.556	0.900**	0.753*	0.605*	0.535
	(1.27)	(2.14)	(1.76)	(1.97)	(1.34)
High	0.679	1.134**	1.033*	0.799**	0.787
	(1.22)	(2.12)	(1.89)	(2.18)	(1.50)
High–Low	0.510	0.579	0.703	0.438	0.595
	(1.10)	(1.24)	(1.47)	(1.01)	(1.21)
Panel B: Composite Earnings Momentum (Orthogonalized)					
Low	0.900***	0.791***	0.878***	0.896***	0.873***
	(4.05)	(3.38)	(3.69)	(3.76)	(3.55)
2	0.968***	0.953***	0.964***	1.080***	1.012***
	(3.77)	(3.59)	(3.54)	(3.97)	(3.60)
3	0.258	0.111	0.143	0.260	0.191
	(1.10)	(0.49)	(0.62)	(1.10)	(0.77)
4	0.511*	0.603**	0.646**	0.783***	0.726**
	(1.80)	(2.37)	(2.48)	(2.86)	(2.55)
High	-0.414	-0.589*	-0.711**	-0.484	-0.583*
	(-1.31)	(-1.94)	(-2.31)	(-1.50)	(-1.77)
High–Low	-1.315***	-1.380***	-1.589***	-1.381***	-1.456***
	(-3.77)	(-3.86)	(-4.45)	(-3.78)	(-3.98)
Panel C: Composite Price- minus Earnings Momentum (Orthogonalized)					
Low	-0.732	-0.236	-0.548	-0.535	-0.680
	(-1.37)	(-0.42)	(-0.98)	(-1.00)	(-1.23)
High	1.093*	1.723***	1.744***	1.283***	1.370**
	(1.93)	(2.94)	(2.90)	(2.72)	(2.34)
High–Low	1.825***	1.959***	2.292***	1.819***	2.050***
	(2.99)	(3.03)	(3.53)	(2.84)	(3.06)

Table 9: Edgar Downloads and Price- and Earnings Momentum

We conduct a two-way sorted portfolio analysis and report the relation between an attention proxy, *Edgar Downloads*, and long–short portfolio returns for the orthogonalized composite momentum portfolios. Each month, we sort stocks into quintiles based on the attention measure, and then within each attention quintile we double-sort stocks into quintiles according to their orthogonalized composite price or earnings momentum measures. Stocks are held for one month and the portfolio returns are equal-weighted. The average long–short orthogonalized momentum portfolio excess returns ($R_p - R_f$) and alphas from the q-4, q-5, Fama and French (2018) six-factor (FF6), and the DHS behavioral factor models are reported for each attention quintile, with the spread between the high- and low-attention quintiles reported as High–Low. Panels A and B report the results for the orthogonalized composite price momentum measure and the orthogonalized composite earnings momentum measure, respectively. Panel C reports the return differences between Panels A and B for the Low, High, and High–Low portfolios. See Tables 1 and 2 for variable descriptions. *, **, and *** denote statistical significance of the percentage abnormal monthly returns at the 10%, 5%, and 1% levels, respectively, with the associated t-statistics in parentheses.

Edgar Downloads Quintile	Long–Short Portfolio Returns				
	$R_p - R_f$	q-4 alpha	q-5 alpha	FF6 alpha	DHS alpha
Panel A: Composite Price Momentum (Orthogonalized)					
Low	-0.446	-0.779***	-0.764**	-0.626**	-0.567*
	(-1.36)	(-2.72)	(-2.55)	(-2.51)	(-1.76)
2	-0.570	-1.057***	-1.058***	-0.898***	-0.972**
	(-1.34)	(-2.93)	(-2.81)	(-2.87)	(-2.47)
3	0.237	0.027	-0.028	0.138	0.183
	(0.64)	(0.08)	(-0.08)	(0.52)	(0.51)
4	0.316	0.066	0.059	0.435*	0.268
	(0.74)	(0.18)	(0.16)	(1.71)	(0.71)
High	0.473	0.313	0.300	0.748***	0.655
	(0.98)	(0.84)	(0.78)	(3.15)	(1.61)
High–Low	0.919***	1.092***	1.064***	1.374***	1.222***
	(2.61)	(3.38)	(3.16)	(4.86)	(3.86)
Panel B: Composite Earnings Momentum (Orthogonalized)					
Low	1.892***	1.871***	1.800***	1.831***	1.894***
	(9.14)	(8.69)	(8.04)	(8.58)	(8.82)
2	1.133***	1.065***	1.023***	1.154***	1.186***
	(5.35)	(4.84)	(4.46)	(5.33)	(5.58)
3	0.503**	0.435**	0.559***	0.510**	0.558***
	(2.57)	(2.14)	(2.66)	(2.48)	(2.72)
4	0.506***	0.490***	0.538***	0.575***	0.602***
	(2.70)	(2.63)	(2.78)	(3.09)	(3.14)
High	0.455**	0.419**	0.511**	0.495**	0.602***
	(2.08)	(2.07)	(2.44)	(2.56)	(2.92)
High–Low	-1.436***	-1.452***	-1.289***	-1.336***	-1.293***
	(-5.04)	(-5.56)	(-4.79)	(-5.26)	(-4.66)
Panel C: Composite Price- minus Earnings Momentum (Orthogonalized)					
Low	-2.338***	-2.651***	-2.564***	-2.457***	-2.461***
	(-5.08)	(-6.16)	(-5.72)	(-6.27)	(-5.33)
High	0.018	-0.106	-0.211	0.253	0.053
	(0.04)	(-0.28)	(-0.54)	(0.87)	(0.13)
High–Low	2.355***	2.544***	2.353***	2.710***	2.514***
	(5.99)	(6.19)	(5.52)	(6.66)	(6.14)

Table 10: Social Attention and Price- and Earnings Momentum

We conduct a two-way sorted portfolio analysis and report the relation between an attention proxy, *Social Attention* (the first principal component of attention from StockTwits, X, and Seeking Alpha provided by Cookson et al. 2024a), and long–short portfolio returns for the orthogonalized composite momentum portfolios. Each month, we sort stocks into quintiles based on the attention measure, and then within each attention quintile we double-sort stocks into quintiles according to their orthogonalized composite price or earnings momentum measures. Stocks are held for one month and the portfolio returns are equal-weighted. The average long–short orthogonalized momentum portfolio excess returns ($R_p - R_f$) and alphas from the q-4, q-5, Fama and French (2018) six-factor (FF6), and the DHS behavioral factor models are reported for each attention quintile, with the spread between the high- and low-attention quintiles reported as High–Low. Panels A and B report the results for the orthogonalized composite price momentum measure and the orthogonalized composite earnings momentum measure, respectively. Panel C reports the return differences between Panels A and B for the Low, High, and High–Low portfolios. See Tables 1 and 2 for variable descriptions. *, **, and *** denote statistical significance of the percentage abnormal monthly returns at the 10%, 5%, and 1% levels, respectively, with the associated t-statistics in parentheses.

Social Attention Quintile	Long–Short Portfolio Returns				
	$R_p - R_f$	q-4 alpha	q-5 alpha	FF6 alpha	DHS alpha
Panel A: Composite Price Momentum (Orthogonalized)					
Low	-0.245	-0.855	-1.051	-0.931	-0.700
	(-0.27)	(-0.94)	(-1.14)	(-1.01)	(-0.76)
2	0.498	0.339	0.154	0.079	-0.051
	(0.63)	(0.42)	(0.19)	(0.10)	(-0.07)
3	2.030***	1.864**	1.566**	1.418**	1.884**
	(2.86)	(2.53)	(2.12)	(2.04)	(2.57)
4	1.600**	1.402*	1.320*	0.881	1.336*
	(2.23)	(1.87)	(1.72)	(1.26)	(1.80)
High	2.092***	2.089***	2.026***	1.620**	1.762***
	(3.29)	(3.08)	(2.93)	(2.51)	(2.71)
High–Low	2.337**	2.944***	3.077***	2.551**	2.462**
	(2.23)	(2.67)	(2.74)	(2.26)	(2.21)
Panel B: Composite Earnings Momentum (Orthogonalized)					
Low	1.460*	1.811**	1.997**	1.805**	1.778**
	(1.91)	(2.23)	(2.42)	(2.19)	(2.19)
2	1.014	1.140*	1.077	1.112*	1.297*
	(1.62)	(1.76)	(1.63)	(1.70)	(1.98)
3	-0.562	-0.435	-0.292	-0.491	-0.396
	(-1.07)	(-0.79)	(-0.53)	(-0.89)	(-0.73)
4	0.347	0.156	0.288	0.292	0.226
	(0.48)	(0.21)	(0.39)	(0.39)	(0.31)
High	-0.340	-0.451	-0.655	-0.321	-0.625
	(-0.48)	(-0.64)	(-0.91)	(-0.44)	(-0.87)
High–Low	-1.800*	-2.262**	-2.652**	-2.126*	-2.404**
	(-1.72)	(-2.10)	(-2.45)	(-1.93)	(-2.20)
Panel C: Composite Price- minus Earnings Momentum (Orthogonalized)					
Low	-1.706	-2.666*	-3.048*	-2.737*	-2.478
	(-1.15)	(-1.72)	(-1.93)	(-1.74)	(-1.59)
High	2.432**	2.541**	2.681**	1.940*	2.387**
	(2.26)	(2.24)	(2.32)	(1.71)	(2.08)
High–Low	4.138**	5.206***	5.729***	4.677**	4.865**
	(2.27)	(2.75)	(2.99)	(2.41)	(2.53)

Table 11: News Articles and Price- and Earnings Momentums

We conduct a two-way sorted portfolio analysis and report the relation between an attention proxy, *News Articles*, and long–short portfolio returns for the orthogonalized composite momentum portfolios. Each month, we sort stocks into quintiles based on the attention measure, and then within each attention quintile we double-sort stocks into quintiles according to their orthogonalized composite price or earnings momentum measures. Stocks are held for one month and the portfolio returns are equal-weighted. The average long–short orthogonalized momentum portfolio excess returns ($R_p - R_f$) and alphas from the q-4, q-5, Fama and French (2018) six-factor (FF6), and the DHS behavioral factor models are reported for each attention quintile, with the spread between the high- and low-attention quintiles reported as High–Low. Panels A and B report the results for the orthogonalized composite price momentum measure and the orthogonalized composite earnings momentum measure, respectively. Panel C reports the return differences between Panels A and B for the Low, High, and High–Low portfolios. See Tables 1 and 2 for variable descriptions. *, **, and *** denote statistical significance of the percentage abnormal monthly returns at the 10%, 5%, and 1% levels, respectively, with the associated t-statistics in parentheses.

News Articles Quintile	Long–Short Portfolio Returns				
	$R_p - R_f$	q-4 alpha	q-5 alpha	FF6 alpha	DHS alpha
Panel A: Composite Price Momentum (Orthogonalized)					
Low	-0.030	-0.307	-0.164	-0.263	-0.176
	(-0.09)	(-0.99)	(-0.52)	(-0.89)	(-0.54)
2	0.496	0.079	0.058	0.021	0.058
	(1.37)	(0.26)	(0.18)	(0.08)	(0.18)
3	0.330	-0.068	-0.175	-0.036	-0.088
	(0.83)	(-0.20)	(-0.50)	(-0.13)	(-0.25)
4	0.542	0.394	0.328	0.357	0.313
	(1.33)	(1.20)	(0.97)	(1.59)	(0.91)
High	0.294	0.031	0.027	-0.062	0.115
	(0.67)	(0.09)	(0.08)	(-0.32)	(0.31)
High–Low	0.324	0.338	0.191	0.201	0.291
	(0.92)	(1.00)	(0.55)	(0.66)	(0.84)
Panel B: Composite Earnings Momentum (Orthogonalized)					
Low	1.660***	1.683***	1.576***	1.636***	1.658***
	(9.65)	(9.39)	(8.62)	(9.04)	(9.18)
2	1.473***	1.444***	1.483***	1.485***	1.537***
	(7.42)	(6.98)	(6.95)	(7.07)	(7.42)
3	1.207***	1.076***	1.136***	1.153***	1.106***
	(6.18)	(5.42)	(5.56)	(5.67)	(5.45)
4	0.435**	0.282	0.415**	0.476**	0.378*
	(2.19)	(1.50)	(2.16)	(2.36)	(1.85)
High	0.156	0.070	0.165	0.240	0.126
	(0.92)	(0.46)	(1.06)	(1.48)	(0.77)
High–Low	-1.504***	-1.614***	-1.411***	-1.396***	-1.532***
	(-6.45)	(-7.36)	(-6.41)	(-6.17)	(-6.56)
Panel C: Composite Price- minus Earnings Momentum (Orthogonalized)					
Low	-1.690***	-1.991***	-1.740***	-1.899***	-1.834***
	(-3.79)	(-4.77)	(-4.09)	(-4.71)	(-4.23)
High	0.138	-0.039	-0.138	-0.302	-0.011
	(0.32)	(-0.10)	(-0.35)	(-1.13)	(-0.03)
High–Low	1.828***	1.951***	1.602***	1.597***	1.823***
	(4.31)	(4.46)	(3.63)	(3.71)	(4.11)

Table 12: Fama-MacBeth Regressions

Each month, from 1971–2022, we estimate Fama and MacBeth (1973) cross-sectional regressions by regressing the month t return of stocks on explanatory variables observed at month $t-1$ and report the time-series average of the coefficients, the average adjusted R-Squared, and the average number of firms per month. The explanatory variables are as follows: The market *Beta* is estimated every year based on the stock's last three years of monthly returns, *Size* is the lagged month market cap, *B/M* is the book-to-market equity ratio, *Lag Return* is the stock's lagged month return, *Idiosyncratic Volatility* is the standard deviation of residuals from daily returns in the last month regressed on the daily Fama-French three factors, *Composite Attention Quintile* is the quintile rank of stocks sorted based on the *Composite Attention* measure, *Composite Price Momentum Score* is the average, scaled between 0 and 1, of the stock's ranks based on the *Past 6-mth Ret* and *Past 11-mth Ret* price momentum measures sorted into percentiles, and *Composite Earnings Momentum Score* is the average, scaled between 0 and 1, of the stock's ranks based on the *SUE*, *ABR*, and *FREV* earnings momentum measures sorted into percentiles. See Tables 1 and 2 for descriptions of the attention and momentum measures. The coefficients are multiplied by 100 and *, **, and *** denote statistical significance of the coefficients at the 10%, 5%, and 1% levels, respectively, with the associated t-statistics in parentheses.

Variables	(1)	(2)	(3)	(4)	(5)
Intercept	2.753***	1.746***	2.161***	1.473**	1.727***
	(4.19)	(2.62)	(3.28)	(2.27)	(2.64)
Beta	0.043	0.056	0.026	0.026	0.026
	(0.74)	(0.84)	(0.43)	(0.43)	(0.43)
Log(Size)	-0.096***	-0.095***	-0.108***	-0.106***	-0.107***
	(-2.84)	(-2.73)	(-3.21)	(-3.18)	(-3.20)
Log(B/M)	0.237***	0.249***	0.245***	0.244***	0.244***
	(4.83)	(5.21)	(5.00)	(4.98)	(5.00)
Lag Return	-5.041***	-5.677***	-5.714***	-5.702***	-5.713***
	(-13.61)	(-14.68)	(-14.72)	(-14.70)	(-14.73)
Idiosyncratic Volatility	-9.304***	-10.517***	-9.896***	-9.890***	-9.986***
	(-3.96)	(-4.09)	(-4.01)	(-4.00)	(-4.04)
Composite Attention Quintile	-0.058	-0.022	-0.102**	0.116**	0.043
	(-1.63)	(-0.58)	(-2.00)	(2.31)	(0.75)
Composite Price Momentum Score	1.111***		-0.165	0.325	-0.504**
	(5.06)		(-0.77)	(1.44)	(-2.28)
Composite Price Momentum Score			0.142**		0.257***
× Composite Attention Quintile			(2.18)		(3.92)
Composite Earnings Momentum Score		2.971***	2.844***	3.690***	4.044***
		(21.82)	(23.93)	(20.33)	(24.39)
Composite Earnings Momentum Score				-0.291***	-0.405***
× Composite Attention Quintile				(-4.94)	(-7.32)
Avg Adj R-Sq	0.050	0.051	0.057	0.056	0.057
Avg # firms/month	3,852	3,601	3,601	3,601	3,601

Figure 1: Cumulative Composite Price Momentum Alphas for High- and Low-Attention Stocks

q-4 alphas are estimated from long–short *Composite Price Momentum (Orthogonalized)* equal-weighted portfolios for the high- and low-*Composite Attention* quintiles. See Tables 1 and 2 for variable descriptions. The alphas are estimated from calendar-time portfolios (1971–2022) that hold stocks in month j after portfolio formation, where $j \in \{1, 2, \ldots, 24\}$. The chart then plots the cumulative q-4 alphas for each attention group from months 1 to 24. The portfolios exclude microcap stocks (the smallest size quintile based on NYSE breakpoints) and momentum-crash months (proxied by the past three-year market buy-and-hold returns being negative).

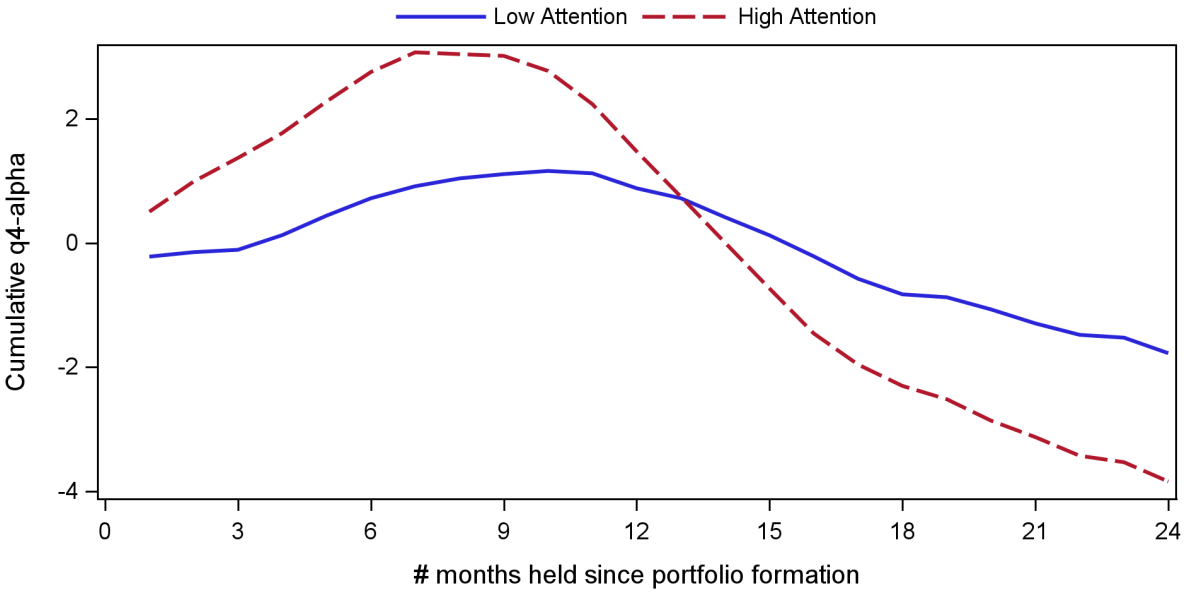
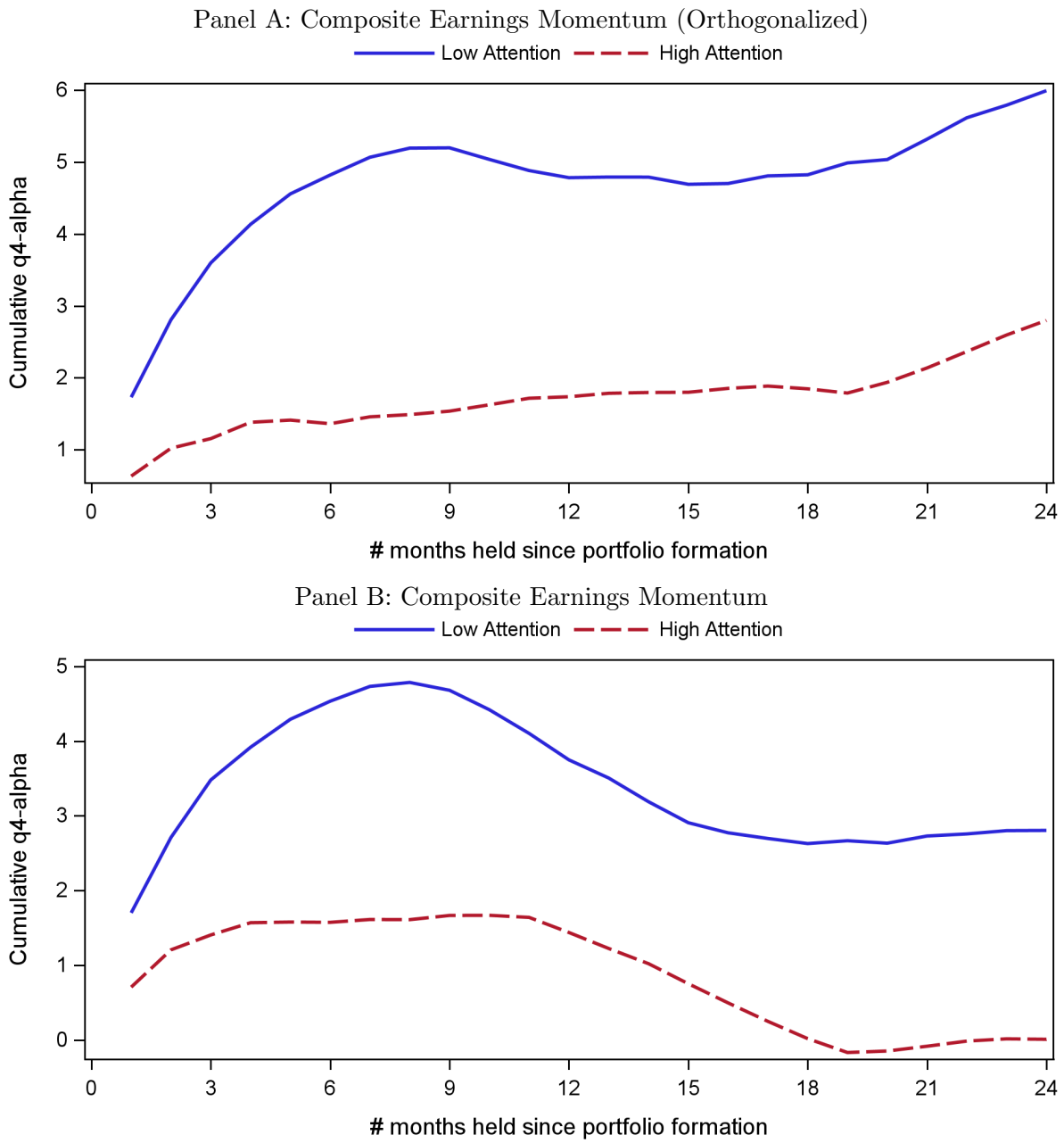


Figure 2: Cumulative Composite Earnings Momentum Alphas for High- and Low-Attention Stocks

q-4 alphas are estimated from long–short earnings momentum equal-weighted portfolios for the high- and low-*Composite Attention* quintiles. Panel A plots results for the *Composite Earnings Momentum (Orthogonalized)* measure and Panel B plots results for the *Composite Earnings Momentum* measure. See Tables 1 and 2 for variable descriptions. The alpha is estimated from calendar-time portfolios (1971–2022) that hold stocks in month j after portfolio formation, where $j \in \{1, 2, \ldots, 24\}$. The chart then plots the cumulative q-4 alphas for each attention group from months 1 to 24.



INTERNET APPENDIX

for

A Tale of Two Anomalies:
The Implications of Investor Attention for Price and Earnings Momentum

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Table IA.1: Composite Attention (3-Month) and Composite Price- and Earnings Momentum (Orthogonalized)

We conduct a two-way sorted portfolio analysis and report the relation between *Composite Attention (3-Month)*, based on the past three months of attention, and long–short portfolio returns for the orthogonalized composite momentum portfolios. Each month, we sort stocks into quintiles based on *Composite Attention (3-Month)*, and then within each attention quintile we double-sort stocks into quintiles according to their orthogonalized composite price or earnings momentum measures. Stocks are held for one month and the portfolio returns are equal-weighted. The average long–short orthogonalized momentum portfolio excess returns ($R_p - R_f$) and alphas from the q-4, q-5, Fama and French (2018) six-factor (FF6), and the DHS behavioral factor models are reported for each attention quintile, with the spread between the high- and low-attention quintiles reported as High–Low. Panels A and B report the results for the orthogonalized composite price momentum measure and the orthogonalized composite earnings momentum measure, respectively. Panel C reports the return differences between Panels A and B for the Low, High, and High–Low portfolios. See Tables 1 and 2 for variable descriptions. *, **, and *** denote statistical significance of the percentage abnormal monthly returns at the 10%, 5%, and 1% levels, respectively, with the associated t-statistics in parentheses.

Composite Attention (3-Month) Quintile	Long–Short Portfolio Returns				
	$R_p - R_f$	q-4 alpha	q-5 alpha	FF6 alpha	DHS alpha
Panel A: Composite Price Momentum (Orthogonalized)					
Low	-0.169	-0.608***	-0.656***	-0.655***	-0.670***
	(-0.90)	(-3.53)	(-3.53)	(-4.22)	(-3.47)
2	0.355*	-0.181	-0.308	-0.307**	-0.281
	(1.69)	(-0.96)	(-1.51)	(-1.97)	(-1.32)
3	0.525**	-0.052	-0.196	-0.234	-0.225
	(2.32)	(-0.25)	(-0.87)	(-1.51)	(-0.98)
4	0.526**	-0.114	-0.296	-0.313**	-0.345
	(2.18)	(-0.53)	(-1.27)	(-1.98)	(-1.44)
High	0.930***	0.280	0.141	-0.039	-0.073
	(3.24)	(1.05)	(0.49)	(-0.22)	(-0.25)
High–Low	1.099***	0.888***	0.796***	0.616***	0.597**
	(4.95)	(3.85)	(3.20)	(3.15)	(2.50)
Panel B: Composite Earnings Momentum (Orthogonalized)					
Low	1.781***	1.693***	1.691***	1.739***	1.716***
	(19.58)	(17.60)	(16.30)	(18.33)	(16.94)
2	1.362***	1.226***	1.251***	1.351***	1.290***
	(14.30)	(12.27)	(11.62)	(13.43)	(12.14)
3	1.243***	1.052***	1.126***	1.198***	1.177***
	(11.57)	(9.62)	(9.57)	(10.75)	(9.81)
4	1.162***	1.015***	1.091***	1.125***	1.064***
	(11.38)	(9.60)	(9.59)	(10.43)	(9.54)
High	1.048***	0.756***	0.681***	0.956***	0.717***
	(7.22)	(5.42)	(4.53)	(6.47)	(4.62)
High–Low	-0.733***	-0.937***	-1.011***	-0.782***	-0.999***
	(-4.76)	(-6.18)	(-6.18)	(-5.01)	(-6.04)
Panel C: Composite Price- minus Earnings Momentum (Orthogonalized)					
Low	-1.950***	-2.301***	-2.347***	-2.394***	-2.386***
	(-8.75)	(-10.64)	(-10.06)	(-12.22)	(-10.19)
High	-0.118	-0.476	-0.540	-0.996***	-0.791**
	(-0.38)	(-1.46)	(-1.54)	(-3.89)	(-2.33)
High–Low	1.832***	1.825***	1.807***	1.398***	1.596***
	(6.65)	(6.21)	(5.70)	(5.06)	(5.19)

Table IA.2: Google ASV and Price- and Earnings Momentum

We conduct a two-way sorted portfolio analysis and report the relation between an attention proxy, *Google ASV*, and long–short portfolio returns for the orthogonalized composite momentum portfolios. Each month, we sort stocks into quintiles based on the attention measure, and then within each attention quintile we double-sort stocks into quintiles according to their orthogonalized composite price or earnings momentum measures. Stocks are held for one month and the portfolio returns are equal-weighted. The average long–short orthogonalized momentum portfolio excess returns ($R_p - R_f$) and alphas from the q-4, q-5, Fama and French (2018) six-factor (FF6), and the DHS behavioral factor models are reported for each attention quintile, with the spread between the high- and low-attention quintiles reported as High–Low. Panels A and B report the results for the orthogonalized composite price momentum measure and the orthogonalized composite earnings momentum measure, respectively. Panel C reports the return differences between Panels A and B for the Low, High, and High–Low portfolios. See Tables 1 and 2 for variable descriptions. *, **, and *** denote statistical significance of the percentage abnormal monthly returns at the 10%, 5%, and 1% levels, respectively, with the associated t-statistics in parentheses.

Google ASV Quintile	Long–Short Portfolio Returns				
	$R_p - R_f$	q-4 alpha	q-5 alpha	FF6 alpha	DHS alpha
Panel A: Composite Price Momentum (Orthogonalized)					
Low	0.485	0.432	0.412	0.523*	0.433
	(1.18)	(1.23)	(1.14)	(1.86)	(1.22)
2	0.272	0.292	0.407	0.382	0.352
	(0.65)	(0.79)	(1.07)	(1.44)	(0.94)
3	0.314	0.241	0.411	0.293	0.362
	(0.75)	(0.63)	(1.05)	(1.04)	(0.93)
4	0.357	0.367	0.385	0.411*	0.445
	(0.87)	(1.05)	(1.07)	(1.66)	(1.25)
High	0.096	0.011	0.121	0.096	0.252
	(0.19)	(0.02)	(0.25)	(0.27)	(0.51)
High–Low	-0.389	-0.421	-0.291	-0.426	-0.182
	(-1.07)	(-1.11)	(-0.75)	(-1.15)	(-0.48)
Panel B: Composite Earnings Momentum (Orthogonalized)					
Low	0.509**	0.446**	0.541**	0.552**	0.547**
	(2.40)	(2.10)	(2.50)	(2.51)	(2.50)
2	0.479**	0.446**	0.464**	0.551**	0.601**
	(2.09)	(2.02)	(2.04)	(2.38)	(2.58)
3	0.527**	0.464**	0.417*	0.509**	0.596**
	(2.35)	(2.05)	(1.80)	(2.25)	(2.60)
4	0.797***	0.670***	0.730***	0.737***	0.802***
	(3.31)	(2.88)	(3.05)	(3.01)	(3.24)
High	0.317	0.091	0.073	0.227	0.241
	(1.00)	(0.33)	(0.26)	(0.77)	(0.79)
High–Low	-0.192	-0.355	-0.469	-0.325	-0.306
	(-0.66)	(-1.27)	(-1.64)	(-1.13)	(-1.05)
Panel C: Composite Price- minus Earnings Momentum (Orthogonalized)					
Low	-0.024	-0.014	-0.129	-0.030	-0.114
	(-0.05)	(-0.03)	(-0.29)	(-0.08)	(-0.26)
High	-0.221	-0.080	0.049	-0.131	0.011
	(-0.39)	(-0.14)	(0.08)	(-0.25)	(0.02)
High–Low	-0.197	-0.066	0.178	-0.101	0.125
	(-0.38)	(-0.13)	(0.33)	(-0.19)	(0.24)

Table IA.3: Google ASV and Price Momentum: Short-term Attention and Subsamples

We conduct a two-way sorted portfolio analysis and report the relation between an attention proxy, *Google ASV*, but measured with abnormal search volume only from the prior month instead of the prior 12 months, and long–short portfolio returns for the orthogonalized composite price momentum portfolios. Each month, we sort stocks into quintiles based on the attention measure and then within each attention quintile we double-sort stocks into quintiles according to their orthogonalized composite price momentum measure. Stocks are held for one month and the portfolio returns are equal-weighted. The portfolios exclude stocks with lagged prices of less than five dollars, microcap stocks (the smallest size quintile based on NYSE breakpoints), and momentum-crash months (proxied by the past three-year market buy-and-hold returns being negative). The average long–short orthogonalized price momentum portfolio excess returns ($R_p - R_f$) and alphas from the q-4, q-5, Fama and French (2018) six-factor (FF6), the DHS behavioral factor, and the Fama and French (1993) three-factor (FF3) models are reported for each attention quintile, with the spread between the high- and low-attention quintiles reported as High–Low. Panels A and B report the results for the 2004–2008 and 2004–2021 sample periods, respectively. See Tables 1 and 2 for variable descriptions. *, **, and *** denote statistical significance of the percentage abnormal monthly returns at the 10%, 5%, and 1% levels, respectively, with the associated t-statistics in parentheses.

Google ASV (Avg from Prior Month) Quintile	Long–Short Portfolio Returns					
	$R_p - R_f$	q-4 alpha	q-5 alpha	FF6 alpha	DHS alpha	FF3 alpha
	Panel A: 2004–2008 sample period					
Low	0.742	-0.409	-0.442	-0.175	0.613	0.640
	(1.31)	(-0.77)	(-0.81)	(-0.40)	(1.17)	(1.14)
2	0.961	-0.332	-0.295	-0.200	0.789	0.947
	(1.39)	(-0.52)	(-0.44)	(-0.42)	(1.47)	(1.48)
3	0.254	-1.095**	-1.052**	-0.564	0.162	0.220
	(0.43)	(-2.29)	(-2.13)	(-1.27)	(0.34)	(0.42)
4	0.351	-0.845*	-0.742	-0.380	0.274	0.249
	(0.64)	(-1.76)	(-1.51)	(-0.89)	(0.61)	(0.50)
High	1.913***	0.799	0.526	1.173**	1.827***	1.885***
	(3.04)	(1.29)	(0.86)	(2.31)	(3.09)	(3.12)
High–Low	1.171**	1.208**	0.968	1.348**	1.214**	1.244**
	(2.29)	(2.08)	(1.68)	(2.35)	(2.27)	(2.44)
	Panel B: 2004–2021 sample period					
Low	0.361	0.369	0.189	0.387	0.355	0.442
	(1.24)	(1.34)	(0.67)	(1.57)	(1.31)	(1.57)
2	0.405	0.472	0.355	0.396	0.382	0.470
	(1.30)	(1.56)	(1.14)	(1.47)	(1.34)	(1.56)
3	0.146	0.190	0.122	0.119	0.142	0.219
	(0.50)	(0.68)	(0.42)	(0.46)	(0.52)	(0.78)
4	0.196	0.346	0.297	0.224	0.253	0.305
	(0.71)	(1.27)	(1.05)	(0.91)	(0.99)	(1.15)
High	0.593*	0.616**	0.347	0.594**	0.624**	0.649**
	(1.91)	(2.10)	(1.18)	(2.22)	(2.15)	(2.20)
High–Low	0.232	0.247	0.158	0.207	0.269	0.207
	(0.96)	(0.98)	(0.61)	(0.81)	(1.06)	(0.83)

Table IA.4: StockTwits and Price- and Earnings Momentum

We conduct a two-way sorted portfolio analysis and report the relation between an attention proxy, *StockTwits* provided by Cookson and Niessner (2020), and long–short portfolio returns for the orthogonalized composite momentum portfolios. Each month, we sort stocks into quintiles based on the attention measure, and then within each attention quintile we double-sort stocks into quintiles according to their orthogonalized composite price or earnings momentum measures. Stocks are held for one month and the portfolio returns are equal-weighted. The average long–short orthogonalized momentum portfolio excess returns ($R_p - R_f$) and alphas from the q-4, q-5, Fama and French (2018) six-factor (FF6), and the DHS behavioral factor models are reported for each attention quintile, with the spread between the high- and low-attention quintiles reported as High–Low. Panels A and B report the results for the orthogonalized composite price momentum measure and the orthogonalized composite earnings momentum measure, respectively. Panel C reports the return differences between Panels A and B for the Low, High, and High–Low portfolios. See Tables 1 and 2 for variable descriptions. *, **, and *** denote statistical significance of the percentage abnormal monthly returns at the 10%, 5%, and 1% levels, respectively, with the associated t-statistics in parentheses.

StockTwits Quintile	Long–Short Portfolio Returns				
	$R_p - R_f$	q-4 alpha	q-5 alpha	FF6 alpha	DHS alpha
Panel A: Composite Price Momentum (Orthogonalized)					
Low	-0.567	-0.258	-0.301	-0.224	-0.087
	(-0.82)	(-0.37)	(-0.41)	(-0.31)	(-0.12)
2	0.442	0.530	0.209	0.436	0.384
	(1.16)	(1.35)	(0.54)	(1.16)	(0.99)
3	-0.195	-0.139	0.104	-0.248	-0.240
	(-0.38)	(-0.30)	(0.22)	(-0.73)	(-0.50)
4	0.431	0.650	0.401	0.520	0.440
	(0.84)	(1.32)	(0.80)	(1.17)	(0.90)
High	1.333***	1.316***	0.960*	1.214***	1.154**
	(2.73)	(2.69)	(1.97)	(2.68)	(2.38)
High–Low	1.900**	1.575**	1.261	1.438*	1.240
	(2.60)	(2.03)	(1.60)	(1.86)	(1.63)
Panel B: Composite Earnings Momentum (Orthogonalized)					
Low	1.835***	1.640***	1.345**	1.610**	1.393**
	(3.14)	(2.68)	(2.17)	(2.60)	(2.26)
2	0.727**	0.692*	0.752**	0.831**	0.809**
	(2.11)	(1.91)	(2.02)	(2.28)	(2.19)
3	0.797*	0.765*	0.434	0.880**	0.715
	(1.86)	(1.74)	(0.99)	(2.16)	(1.60)
4	0.630*	0.566*	0.599*	0.656*	0.653*
	(1.94)	(1.69)	(1.74)	(1.89)	(1.90)
High	0.580	0.653*	0.619	0.694*	0.625*
	(1.60)	(1.79)	(1.65)	(1.87)	(1.69)
High–Low	-1.255*	-0.987	-0.726	-0.916	-0.768
	(-1.80)	(-1.39)	(-1.00)	(-1.26)	(-1.05)
Panel C: Composite Price- minus Earnings Momentum (Orthogonalized)					
Low	-2.402**	-1.899	-1.646	-1.834	-1.480
	(-1.98)	(-1.51)	(-1.28)	(-1.44)	(-1.16)
High	0.753	0.663	0.341	0.520	0.528
	(1.09)	(0.91)	(0.46)	(0.73)	(0.72)
High–Low	3.155**	2.562*	1.987	2.353	2.008
	(2.35)	(1.82)	(1.39)	(1.65)	(1.42)

Table IA.5: Tweets and Price- and Earnings Momentum

We conduct a two-way sorted portfolio analysis and report the relation between an attention proxy, *Tweets*, which is the sum of tweets and retweets on X (formerly Twitter) for the stock provided by Chen and Hwang (2022), and long–short portfolio returns for the orthogonalized composite momentum portfolios. Each month, we sort stocks into quintiles based on the attention measure, and then within each attention quintile we double-sort stocks into quintiles according to their orthogonalized composite price or earnings momentum measures. Stocks are held for one month and the portfolio returns are equal-weighted. The average long–short orthogonalized momentum portfolio excess returns ($R_p - R_f$) and alphas from the q-4, q-5, Fama and French (2018) six-factor (FF6), and the DHS behavioral factor models are reported for each attention quintile, with the spread between the high- and low-attention quintiles reported as High–Low. Panels A and B report the results for the orthogonalized composite price momentum measure and the orthogonalized composite earnings momentum measure, respectively. Panel C reports the return differences between Panels A and B for the Low, High, and High–Low portfolios. See Tables 1 and 2 for variable descriptions. *, **, and *** denote statistical significance of the percentage abnormal monthly returns at the 10%, 5%, and 1% levels, respectively, with the associated t-statistics in parentheses.

Tweets Quintile	Long–Short Portfolio Returns				
	$R_p - R_f$	q-4 alpha	q-5 alpha	FF6 alpha	DHS alpha
Panel A: Composite Price Momentum (Orthogonalized)					
Low	0.040	0.055	-0.309	0.117	-0.044
	(0.09)	(0.13)	(-0.73)	(0.27)	(-0.10)
2	0.279	0.198	-0.106	0.164	0.084
	(0.64)	(0.48)	(-0.26)	(0.47)	(0.21)
3	0.360	0.464	0.117	0.386	0.258
	(0.74)	(1.04)	(0.27)	(1.06)	(0.65)
4	1.221**	1.149**	0.835	1.015**	0.983**
	(2.25)	(2.24)	(1.62)	(2.54)	(2.05)
High	1.294**	1.192*	0.874	0.960**	0.881
	(2.02)	(1.94)	(1.40)	(2.09)	(1.60)
High–Low	1.254**	1.136*	1.183*	0.843	0.924
	(2.16)	(1.89)	(1.89)	(1.54)	(1.56)
Panel B: Composite Earnings Momentum (Orthogonalized)					
Low	1.303***	1.230***	1.300***	1.281***	1.214***
	(7.55)	(6.44)	(6.61)	(6.67)	(6.36)
2	0.789***	1.037***	1.094***	1.027***	0.954***
	(3.55)	(4.47)	(4.56)	(4.51)	(3.94)
3	0.697***	0.573**	0.500**	0.632**	0.670**
	(2.90)	(2.36)	(2.00)	(2.57)	(2.58)
4	0.368	0.257	0.251	0.386	0.422
	(1.32)	(0.84)	(0.79)	(1.24)	(1.35)
High	-0.276	-0.260	-0.293	-0.222	-0.356
	(-0.89)	(-0.79)	(-0.86)	(-0.68)	(-1.11)
High–Low	-1.579***	-1.490***	-1.593***	-1.503***	-1.570***
	(-4.71)	(-4.10)	(-4.24)	(-4.17)	(-4.35)
Panel C: Composite Price- minus Earnings Momentum (Orthogonalized)					
Low	-1.262**	-1.175**	-1.608***	-1.164**	-1.257**
	(-2.63)	(-2.36)	(-3.35)	(-2.41)	(-2.50)
High	1.570**	1.451*	1.168	1.182*	1.237*
	(2.16)	(1.95)	(1.53)	(1.79)	(1.72)
High–Low	2.832***	2.626***	2.776***	2.346***	2.494***
	(4.04)	(3.56)	(3.63)	(3.25)	(3.34)

Table IA.6: –Distraction and Price- and Earnings Momentum

We conduct a two-way sorted portfolio analysis and report the relation between an attention proxy, *–Distraction*, and long–short portfolio returns for the orthogonalized composite momentum portfolios. Each month, we sort stocks into quintiles based on the attention measure, and then within each attention quintile we double-sort stocks into quintiles according to their orthogonalized composite price or earnings momentum measures. Stocks are held for one month and the portfolio returns are equal-weighted. The average long–short orthogonalized momentum portfolio excess returns ($R_p - R_f$) and alphas from the q-4, q-5, Fama and French (2018) six-factor (FF6), and the DHS behavioral factor models are reported for each attention quintile, with the spread between the high- and low-attention quintiles reported as High–Low. Panels A and B report the results for the orthogonalized composite price momentum measure and the orthogonalized composite earnings momentum measure, respectively. Panel C reports the return differences between Panels A and B for the Low, High, and High–Low portfolios. See Tables 1 and 2 for variable descriptions. *, **, and *** denote statistical significance of the percentage abnormal monthly returns at the 10%, 5%, and 1% levels, respectively, with the associated t-statistics in parentheses.

−Distraction Quintile	Long–Short Portfolio Returns				
	$R_p - R_f$	q-4 alpha	q-5 alpha	FF6 alpha	DHS alpha
Panel A: Composite Price Momentum (Orthogonalized)					
Low	0.577**	0.041	-0.088	-0.041	0.006
	(2.41)	(0.19)	(-0.38)	(-0.26)	(0.03)
2	0.346	-0.098	-0.036	-0.166	-0.102
	(1.51)	(-0.46)	(-0.16)	(-1.06)	(-0.44)
3	0.426*	-0.064	-0.055	-0.170	-0.115
	(1.87)	(-0.30)	(-0.24)	(-1.04)	(-0.49)
4	0.603**	0.018	-0.034	-0.100	-0.043
	(2.29)	(0.07)	(-0.13)	(-0.51)	(-0.16)
High	0.355	-0.373	-0.551*	-0.453**	-0.471
	(1.22)	(-1.36)	(-1.89)	(-2.11)	(-1.58)
High–Low	-0.223	-0.415**	-0.463**	-0.412**	-0.477**
	(-1.18)	(-2.06)	(-2.16)	(-2.06)	(-2.31)
Panel B: Composite Earnings Momentum (Orthogonalized)					
Low	1.330***	1.117***	1.121***	1.239***	1.130***
	(11.73)	(10.12)	(9.55)	(10.81)	(9.18)
2	1.027***	0.847***	0.849***	0.942***	0.886***
	(10.32)	(8.57)	(8.08)	(9.27)	(8.15)
3	1.475***	1.282***	1.294***	1.390***	1.388***
	(14.31)	(11.94)	(11.32)	(12.78)	(12.25)
4	1.354***	1.182***	1.247***	1.279***	1.206***
	(11.92)	(10.00)	(9.93)	(10.65)	(9.75)
High	1.306***	1.088***	1.119***	1.213***	1.192***
	(9.80)	(7.88)	(7.62)	(8.63)	(8.16)
High–Low	-0.025	-0.029	-0.002	-0.026	0.063
	(-0.17)	(-0.19)	(-0.01)	(-0.17)	(0.39)
Panel C: Composite Price- minus Earnings Momentum (Orthogonalized)					
Low	-0.753***	-1.076***	-1.209***	-1.280***	-1.123***
	(-2.84)	(-4.04)	(-4.27)	(-5.92)	(-4.03)
High	-0.951***	-1.461***	-1.670***	-1.666***	-1.663***
	(-2.92)	(-4.43)	(-4.78)	(-6.17)	(-4.83)
High–Low	-0.198	-0.385	-0.461*	-0.386	-0.540**
	(-0.80)	(-1.47)	(-1.65)	(-1.49)	(-2.00)

Table IA.7: Independently Sorted Composite Attention and Composite Price- and Earnings Momentum (Orthogonalized)

We conduct an independently sorted portfolio analysis and report the relation between *Composite Attention* and long–short portfolio returns for the orthogonalized composite momentum portfolios. Each month, we sort stocks into quintiles based on *Composite Attention*. We also independently sort stocks according to their orthogonalized composite price or earnings momentum measures. Stocks are held for one month and the portfolio returns are equal-weighted. The average long–short orthogonalized momentum portfolio excess returns ($R_p - R_f$) and alphas from the q-4, q-5, Fama and French (2018) six-factor (FF6), and the DHS behavioral factor models are reported for each attention quintile, with the spread between the high- and low-attention quintiles reported as High–Low. Panels A and B report the results for the orthogonalized composite price momentum measure and the orthogonalized composite earnings momentum measure, respectively. Panel C reports the return differences between Panels A and B for the Low, High, and High–Low portfolios. See Tables 1 and 2 for variable descriptions. *, **, and *** denote statistical significance of the percentage abnormal monthly returns at the 10%, 5%, and 1% levels, respectively, with the associated t-statistics in parentheses.

Composite Attention Quintile	Long–Short Portfolio Returns				
	$R_p - R_f$	q-4 alpha	q-5 alpha	FF6 alpha	DHS alpha
Panel A: Composite Price Momentum (Orthogonalized)					
Low	-0.132	-0.582***	-0.616***	-0.676***	-0.653***
	(-0.64)	(-2.97)	(-2.91)	(-3.84)	(-3.00)
2	0.416*	-0.143	-0.177	-0.258	-0.189
	(1.94)	(-0.71)	(-0.81)	(-1.55)	(-0.85)
3	0.492**	-0.113	-0.199	-0.302*	-0.299
	(2.18)	(-0.54)	(-0.88)	(-1.95)	(-1.30)
4	0.556**	-0.079	-0.331	-0.282*	-0.313
	(2.20)	(-0.35)	(-1.36)	(-1.68)	(-1.24)
High	0.973***	0.334	0.115	0.086	-0.019
	(3.59)	(1.34)	(0.43)	(0.53)	(-0.07)
High-–Low	1.105***	0.916***	0.730***	0.762***	0.634***
	(5.12)	(4.06)	(3.01)	(3.81)	(2.74)
Panel B: Composite Earnings Momentum (Orthogonalized)					
Low	1.775***	1.658***	1.681***	1.724***	1.732***
	(19.26)	(17.09)	(16.07)	(17.81)	(16.89)
2	1.383***	1.286***	1.344***	1.385***	1.267***
	(14.25)	(12.67)	(12.30)	(13.68)	(12.07)
3	1.279***	1.176***	1.174***	1.294***	1.205***
	(12.29)	(10.88)	(10.07)	(11.83)	(10.48)
4	1.111***	0.909***	1.000***	1.078***	1.036***
	(10.03)	(8.10)	(8.29)	(9.32)	(8.45)
High	0.953***	0.622***	0.637***	0.797***	0.669***
	(7.21)	(4.91)	(4.67)	(5.92)	(4.74)
High–Low	-0.822***	-1.037***	-1.044***	-0.928***	-1.063***
	(-5.58)	(-7.01)	(-6.54)	(-6.15)	(-6.73)
Panel C: Composite Price- minus Earnings Momentum (Orthogonalized)					
Low	-1.907***	-2.241***	-2.297***	-2.400***	-2.385***
	(-7.83)	(-9.24)	(-8.78)	(-10.89)	(-9.22)
High	0.020	-0.288	-0.523*	-0.711***	-0.688**
	(0.07)	(-1.01)	(-1.71)	(-3.38)	(-2.35)
High–Low	1.927***	1.953***	1.774***	1.689***	1.697***
	(7.31)	(6.92)	(5.84)	(6.30)	(5.83)

Table IA.8: Attention and Composite Price Momentum (Orthogonalized): Influence of Momentum Crashes

We conduct a two-way sorted portfolio analysis and report the relation between attention and long–short portfolio returns for the orthogonalized composite price momentum portfolios. Each month, we first sort stocks into quintiles based on attention, and then within each attention quintile we double-sort stocks into quintiles according to their orthogonalized composite price momentum measure. Stocks are then held for one month and the portfolio returns are equal-weighted. The average long–short orthogonalized price momentum portfolio excess returns ($R_p - R_f$) and alphas from the q-4, q-5, Fama and French (2018) six-factor (FF6), the DHS behavioral factor, and the Fama and French (1993) three-factor (FF3) models are reported for each attention quintile, with the spread between the high- and low-attention quintiles reported as High–Low. Panel A and B both use the *Composite Attention* measure where Panel A reports the results for the full sample and Panel B reporting results that exclude momentum-crash months (proxied by the past three-year market buy-and-hold returns being negative). Panel C reports results that use the *–Distraction* measure excluding momentum-crash months. See Tables 1 and 2 for variable descriptions. *, **, and *** denote statistical significance of the percentage abnormal monthly returns at the 10%, 5%, and 1% levels, respectively, with the associated t-statistics in parentheses.

Composite Attention Quintile	Long–Short Portfolio Returns					
	$R_p - R_f$	q-4 alpha	q-5 alpha	FF6 alpha	DHS alpha	FF3 alpha
Panel A: Composite Price Momentum (Orthogonalized), Full Sample						
Low	-0.163	-0.636***	-0.679***	-0.683***	-0.683***	0.002
	(-0.85)	(-3.62)	(-3.59)	(-4.28)	(-3.42)	(0.01)
2	0.335*	-0.155	-0.169	-0.255	-0.232	0.515**
	(1.65)	(-0.81)	(-0.82)	(-1.62)	(-1.10)	(2.57)
3	0.505**	-0.095	-0.164	-0.277*	-0.285	0.698***
	(2.28)	(-0.46)	(-0.75)	(-1.83)	(-1.26)	(3.17)
4	0.534**	-0.134	-0.361	-0.337**	-0.356	0.775***
	(2.12)	(-0.59)	(-1.49)	(-1.97)	(-1.41)	(3.16)
High	0.912***	0.268	0.095	-0.007	-0.045	1.211***
	(3.28)	(1.03)	(0.34)	(-0.04)	(-0.16)	(4.40)
High–Low	1.076***	0.905***	0.774***	0.676***	0.638***	1.208***
	(4.92)	(3.95)	(3.14)	(3.48)	(2.73)	(5.51)
Panel B: Composite Price Momentum (Orthogonalized), Momentum-Crash Months Excluded						
Low	0.056	-0.325**	-0.472***	-0.338**	-0.328*	0.120
	(0.36)	(-2.10)	(-2.84)	(-2.29)	(-1.95)	(0.76)
2	0.487***	0.085	-0.042	0.045	0.027	0.572***
	(2.81)	(0.49)	(-0.23)	(0.30)	(0.14)	(3.29)
3	0.673***	0.203	0.022	0.080	0.047	0.756***
	(3.57)	(1.08)	(0.11)	(0.50)	(0.24)	(3.97)
4	0.729***	0.182	-0.172	0.042	0.011	0.839***
	(3.29)	(0.84)	(-0.74)	(0.23)	(0.05)	(3.81)
High	1.123***	0.657***	0.336	0.458**	0.409*	1.246***
	(4.92)	(2.82)	(1.35)	(2.52)	(1.69)	(5.40)
High–Low	1.067***	0.981***	0.809***	0.795***	0.737***	1.126***
	(5.66)	(4.88)	(3.74)	(4.50)	(3.62)	(5.91)

Table IA.8: (Cont'd)

–Distraction Quintile	Long–Short Portfolio Returns					
	$R_p - R_f$	q-4 alpha	q-5 alpha	FF6 alpha	DHS alpha	FF3 alpha
Panel C: –Distraction, Momentum-Crash Months Excluded						
Low	0.674***	0.253	-0.002	0.219	0.266	0.801***
	(3.22)	(1.22)	(-0.01)	(1.24)	(1.21)	(3.81)
2	0.484***	0.153	0.059	0.157	0.197	0.620***
	(2.69)	(0.85)	(0.31)	(1.02)	(1.04)	(3.45)
3	0.581***	0.183	0.057	0.136	0.157	0.699***
	(2.93)	(0.92)	(0.27)	(0.79)	(0.75)	(3.51)
4	0.722***	0.222	0.030	0.190	0.174	0.818***
	(3.14)	(0.97)	(0.12)	(0.95)	(0.73)	(3.52)
High	0.583**	0.003	-0.297	-0.017	-0.063	0.666**
	(2.30)	(0.01)	(-1.10)	(-0.08)	(-0.24)	(2.59)
High–Low	-0.092	-0.250	-0.295	-0.237	-0.328*	-0.134
	(-0.53)	(-1.37)	(-1.51)	(-1.30)	(-1.76)	(-0.77)

Table IA.9: Event-Month Composite Price Momentum Alphas for High- and Low-Attention Stocks

q-4 alphas are estimated from long–short *Composite Price Momentum (Orthogonalized)* equal-weighted portfolios for the High and Low *Composite Attention* quintiles. See Tables 1 and 2 for variable descriptions. The alphas are estimated from calendar-time portfolios (1971–2022) that hold stocks in month j after portfolio formation, where $j \in \{1, 2, \ldots, 24\}$. Microcap stocks (the smallest size quintile defined using NYSE breakpoints) and momentum-crash months (proxied by the past three-year market buy-and-hold returns being negative) are excluded. *, **, and *** denote statistical significance of the alphas at the 10%, 5%, and 1% levels, respectively, with the associated t-statistics in parentheses.

Event-month	Composite Price Momentum (Orthogonalized)		
	High Attention	Low Attention	High–Low
1	0.502**	-0.221	0.723***
	(2.01)	(-1.64)	(3.55)
2	0.486*	0.072	0.414*
	(1.91)	(0.56)	(1.95)
3	0.380	0.038	0.342
	(1.57)	(0.30)	(1.63)
4	0.400	0.236*	0.164
	(1.64)	(1.96)	(0.78)
5	0.508**	0.314***	0.195
	(2.17)	(2.59)	(0.95)
6	0.474**	0.279**	0.195
	(2.15)	(2.32)	(1.03)
7	0.317	0.194*	0.123
	(1.42)	(1.66)	(0.65)
8	-0.029	0.126	-0.155
	(-0.13)	(1.08)	(-0.79)
9	-0.028	0.069	-0.097
	(-0.13)	(0.60)	(-0.50)
10	-0.242	0.051	-0.293
	(-1.17)	(0.40)	(-1.56)
11	-0.534**	-0.038	-0.496**
	(-2.46)	(-0.31)	(-2.56)
12	-0.761***	-0.241**	-0.519***
	(-3.50)	(-2.00)	(-2.67)
13	-0.731***	-0.161	-0.570***
	(-3.40)	(-1.29)	(-3.01)
14	-0.742***	-0.306**	-0.437**
	(-3.55)	(-2.49)	(-2.35)
15	-0.735***	-0.290**	-0.445**
	(-3.59)	(-2.41)	(-2.40)
16	-0.724***	-0.339***	-0.385*
	(-3.64)	(-2.82)	(-1.95)
17	-0.500***	-0.361***	-0.139
	(-2.69)	(-3.38)	(-0.76)
18	-0.343*	-0.250**	-0.093
	(-1.84)	(-2.17)	(-0.51)
19	-0.214	-0.047	-0.167
	(-1.11)	(-0.42)	(-0.88)
20	-0.345*	-0.196*	-0.148
	(-1.76)	(-1.75)	(-0.78)
21	-0.265	-0.225*	-0.040
	(-1.32)	(-1.96)	(-0.21)
22	-0.300	-0.184*	-0.116
	(-1.53)	(-1.71)	(-0.62)
23	-0.104	-0.045	-0.058
	(-0.58)	(-0.43)	(-0.32)
24	-0.310	-0.250**	-0.060
	(-1.63)	(-2.20)	(-0.30)

Table IA.10: Event-Month Composite Earnings Momentum Alphas for High- and Low-Attention Stocks

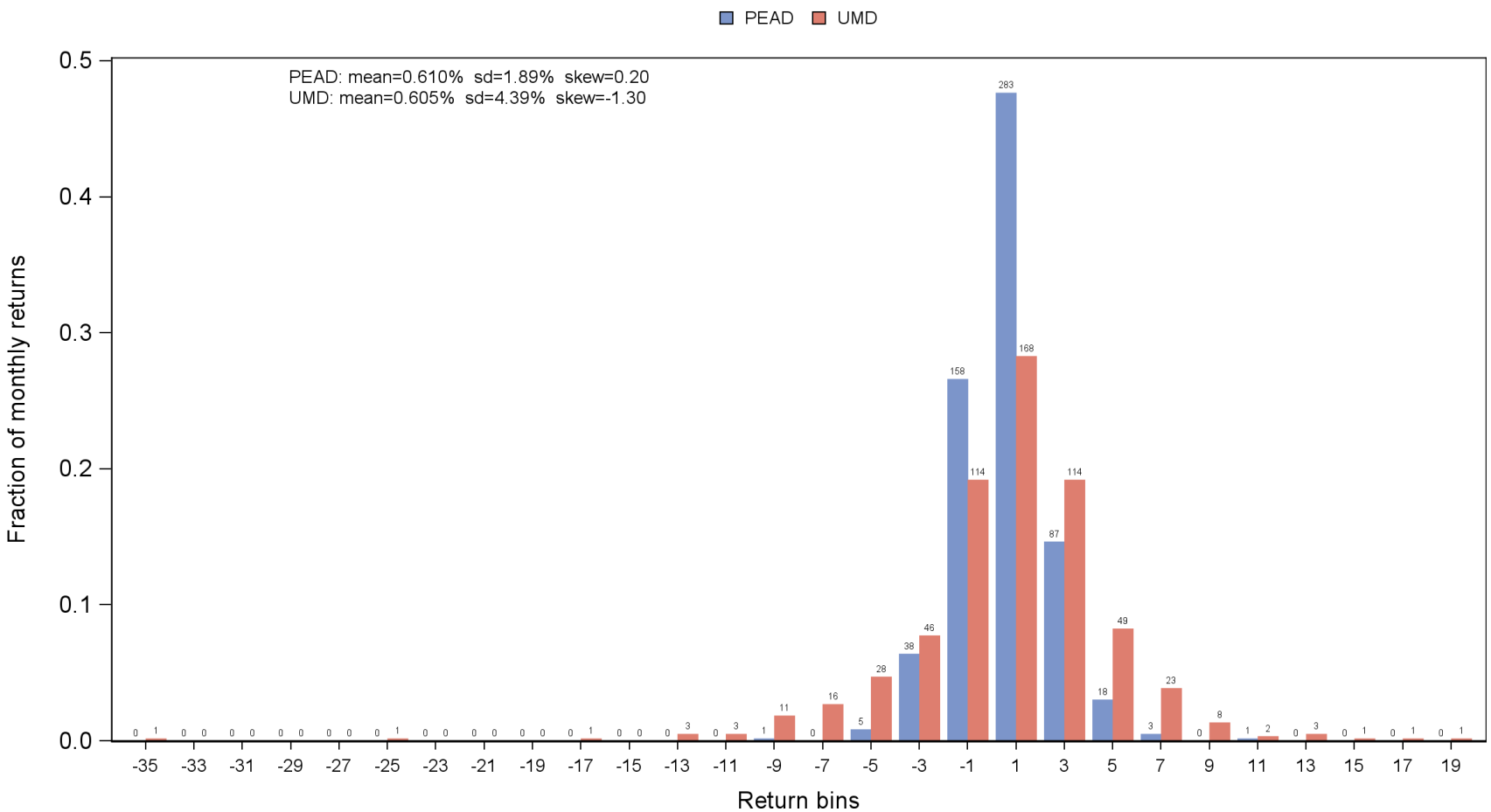
q-4 alphas are estimated from long–short earnings momentum equal-weighted portfolios for the High and Low *Composite Attention* quintiles. The first set of columns uses the *Composite Earnings Momentum (Orthogonalized)* measure, and the second set of columns uses the *Composite Earnings Momentum* measure. See Tables 1 and 2 for variable descriptions. The alphas are estimated from calendar-time portfolios (1971–2022) that hold stocks in month j after portfolio formation, where $j \in \{1, 2, \ldots, 24\}$. *, **, and *** denote statistical significance of the alphas at the 10%, 5%, and 1% levels, respectively, with the associated t-statistics in parentheses.

Event-month	Composite Earnings Momentum (Orthogonalized)			Composite Earnings Momentum		
	High Attention	Low Attention	High–Low	High Attention	Low Attention	High–Low
1	0.634***	1.730***	-1.096***	0.708***	1.702***	-0.994***
	(4.97)	(17.61)	(-7.17)	(4.46)	(17.30)	(-5.92)
2	0.388***	1.077***	-0.689***	0.499***	1.009***	-0.510***
	(3.17)	(11.62)	(-4.71)	(3.31)	(10.11)	(-3.15)
3	0.133	0.790***	-0.657***	0.199	0.769***	-0.570***
	(1.22)	(8.15)	(-4.71)	(1.38)	(7.17)	(-3.62)
4	0.227	0.536***	-0.310**	0.164	0.438***	-0.274
	(1.63)	(5.55)	(-1.97)	(0.99)	(4.04)	(-1.51)
5	0.030	0.423***	-0.393***	0.009	0.374***	-0.365**
	(0.26)	(4.31)	(-2.72)	(0.06)	(3.52)	(-2.19)
6	-0.050	0.262***	-0.312**	-0.004	0.242**	-0.246
	(-0.43)	(2.75)	(-2.16)	(-0.03)	(2.41)	(-1.51)
7	0.096	0.247***	-0.152	0.038	0.198**	-0.160
	(0.85)	(2.74)	(-1.09)	(0.28)	(2.09)	(-1.07)
8	0.031	0.127	-0.096	-0.002	0.053	-0.055
	(0.26)	(1.38)	(-0.67)	(-0.02)	(0.54)	(-0.36)
9	0.048	0.004	0.044	0.056	-0.105	0.161
	(0.42)	(0.04)	(0.30)	(0.43)	(-1.02)	(1.01)
10	0.088	-0.161	0.249*	0.002	-0.260**	0.262
	(0.81)	(-1.62)	(1.67)	(0.01)	(-2.45)	(1.64)
11	0.091	-0.155	0.246*	-0.028	-0.317***	0.289*
	(0.80)	(-1.61)	(1.73)	(-0.22)	(-3.14)	(1.94)
12	0.021	-0.099	0.120	-0.202	-0.354***	0.152
	(0.18)	(-1.08)	(0.87)	(-1.60)	(-3.22)	(1.01)
13	0.048	0.009	0.039	-0.214*	-0.243**	0.029
	(0.41)	(0.09)	(0.28)	(-1.66)	(-2.16)	(0.19)
14	0.011	-0.001	0.012	-0.203	-0.317***	0.114
	(0.10)	(-0.01)	(0.09)	(-1.52)	(-2.83)	(0.73)
15	0.003	-0.100	0.102	-0.268**	-0.281***	0.013
	(0.02)	(-0.99)	(0.74)	(-2.18)	(-2.67)	(0.09)
16	0.054	0.011	0.044	-0.257**	-0.135	-0.122
	(0.50)	(0.11)	(0.31)	(-2.06)	(-1.29)	(-0.79)
17	0.032	0.106	-0.074	-0.247**	-0.076	-0.171
	(0.29)	(1.14)	(-0.54)	(-2.07)	(-0.76)	(-1.24)
18	-0.038	0.015	-0.053	-0.231**	-0.068	-0.163
	(-0.33)	(0.15)	(-0.36)	(-1.99)	(-0.64)	(-1.15)
19	-0.059	0.165*	-0.225	-0.186	0.038	-0.224
	(-0.53)	(1.77)	(-1.62)	(-1.60)	(0.37)	(-1.54)
20	0.149	0.047	0.102	0.020	-0.033	0.052
	(1.35)	(0.50)	(0.74)	(0.16)	(-0.31)	(0.35)
21	0.200*	0.283***	-0.083	0.064	0.096	-0.032
	(1.76)	(2.72)	(-0.55)	(0.53)	(0.90)	(-0.21)
22	0.228*	0.298***	-0.070	0.069	0.028	0.041
	(1.79)	(2.91)	(-0.47)	(0.51)	(0.27)	(0.25)
23	0.230*	0.175*	0.055	0.030	0.044	-0.014
	(1.80)	(1.85)	(0.36)	(0.23)	(0.40)	(-0.09)
24	0.203	0.201**	0.002	-0.006	0.003	-0.009
	(1.50)	(2.19)	(0.01)	(-0.04)	(0.03)	(-0.05)

Table IA.11: Composite Attention (Size-Controlled) and Composite Price- and Earnings Momentum

We conduct a two-way sorted portfolio analysis and report the relation between *Composite Attention (Size-Controlled)*, where each of the component attention measures controls for the lagged market cap, and long–short portfolio returns for the orthogonalized composite momentum portfolios. Each month, we sort stocks into quintiles based on the attention measure, and then within each attention quintile we double-sort stocks into quintiles according to their orthogonalized composite price or earnings momentum measures. Stocks are held for one month and the portfolio returns are equal-weighted. The average long–short orthogonalized momentum portfolio excess returns ($R_p - R_f$) and alphas from the q-4, q-5, Fama and French (2018) six-factor (FF6), and the DHS behavioral factor models are reported for each attention quintile, with the spread between the high- and low-attention quintiles reported as High–Low. Panels A and B report the results for the orthogonalized composite price momentum measure and the orthogonalized composite earnings momentum measure, respectively. Panel C reports the return differences between Panels A and B for the Low, High, and High–Low portfolios. See Tables 1 and 2 for variable descriptions. *, **, and *** denote statistical significance of the percentage abnormal monthly returns at the 10%, 5%, and 1% levels, respectively, with the associated t-statistics in parentheses.

Composite Attention (Size-Controlled) Quintile	Long–Short Portfolio Returns				
	$R_p - R_f$	q-4 alpha	q-5 alpha	FF6 alpha	DHS alpha
Panel A: Composite Price Momentum (Orthogonalized)					
Low	-0.210	-0.664***	-0.739***	-0.746***	-0.740***
	(-1.15)	(-3.98)	(-4.11)	(-5.14)	(-3.90)
2	0.254	-0.159	-0.187	-0.276*	-0.271
	(1.38)	(-0.91)	(-1.00)	(-1.96)	(-1.43)
3	0.554***	0.011	-0.129	-0.155	-0.169
	(2.60)	(0.05)	(-0.61)	(-1.05)	(-0.77)
4	0.374	-0.374*	-0.620***	-0.578***	-0.544**
	(1.52)	(-1.68)	(-2.61)	(-3.50)	(-2.18)
High	0.902***	0.247	0.074	-0.002	-0.048
	(3.33)	(0.96)	(0.27)	(-0.01)	(-0.17)
High–Low	1.112***	0.911***	0.813***	0.744***	0.692***
	(5.39)	(4.20)	(3.48)	(3.88)	(3.08)
Panel B: Composite Earnings Momentum (Orthogonalized)					
Low	1.580***	1.463***	1.514***	1.541***	1.541***
	(18.83)	(16.65)	(16.00)	(17.60)	(16.72)
2	1.343***	1.232***	1.236***	1.302***	1.237***
	(14.64)	(12.88)	(11.98)	(13.60)	(12.14)
3	1.286***	1.122***	1.178***	1.257***	1.194***
	(13.38)	(11.30)	(11.02)	(12.49)	(11.16)
4	1.195***	1.062***	1.151***	1.204***	1.133***
	(10.85)	(9.47)	(9.54)	(10.52)	(9.35)
High	1.056***	0.766***	0.760***	0.935***	0.840***
	(7.87)	(5.80)	(5.34)	(6.78)	(5.75)
High–Low	-0.524***	-0.697***	-0.754***	-0.607***	-0.701***
	(-3.59)	(-4.73)	(-4.75)	(-4.05)	(-4.43)
Panel C: Composite Price- minus Earnings Momentum (Orthogonalized)					
Low	-1.790***	-2.127***	-2.253***	-2.288***	-2.282***
	(-8.35)	(-10.31)	(-10.14)	(-12.58)	(-10.24)
High	-0.154	-0.519	-0.685**	-0.936***	-0.888***
	(-0.50)	(-1.64)	(-2.02)	(-3.73)	(-2.72)
High–Low	1.636***	1.608***	1.567***	1.351***	1.393***
	(6.06)	(5.62)	(5.08)	(4.98)	(4.65)

Figure IA.1: Histogram of PEAD Factor versus UMD Factor

The post-earnings announcement drift (PEAD) factor's monthly returns are compared with the up-minus-down (UMD) factor's monthly returns in a histogram with 2% bins, and the mean, standard deviation (sd), and skewness (skew) of the factor returns are reported. The PEAD factor is the one used by Daniel, Hirshleifer, and Sun (2020) (from July 1972 to December 2022, available from the authors' website) and the UMD factor is obtained from Ken French's website. The number on top of each histogram bar indicates the number of months that a return (%) that falls into the return bin is observed.